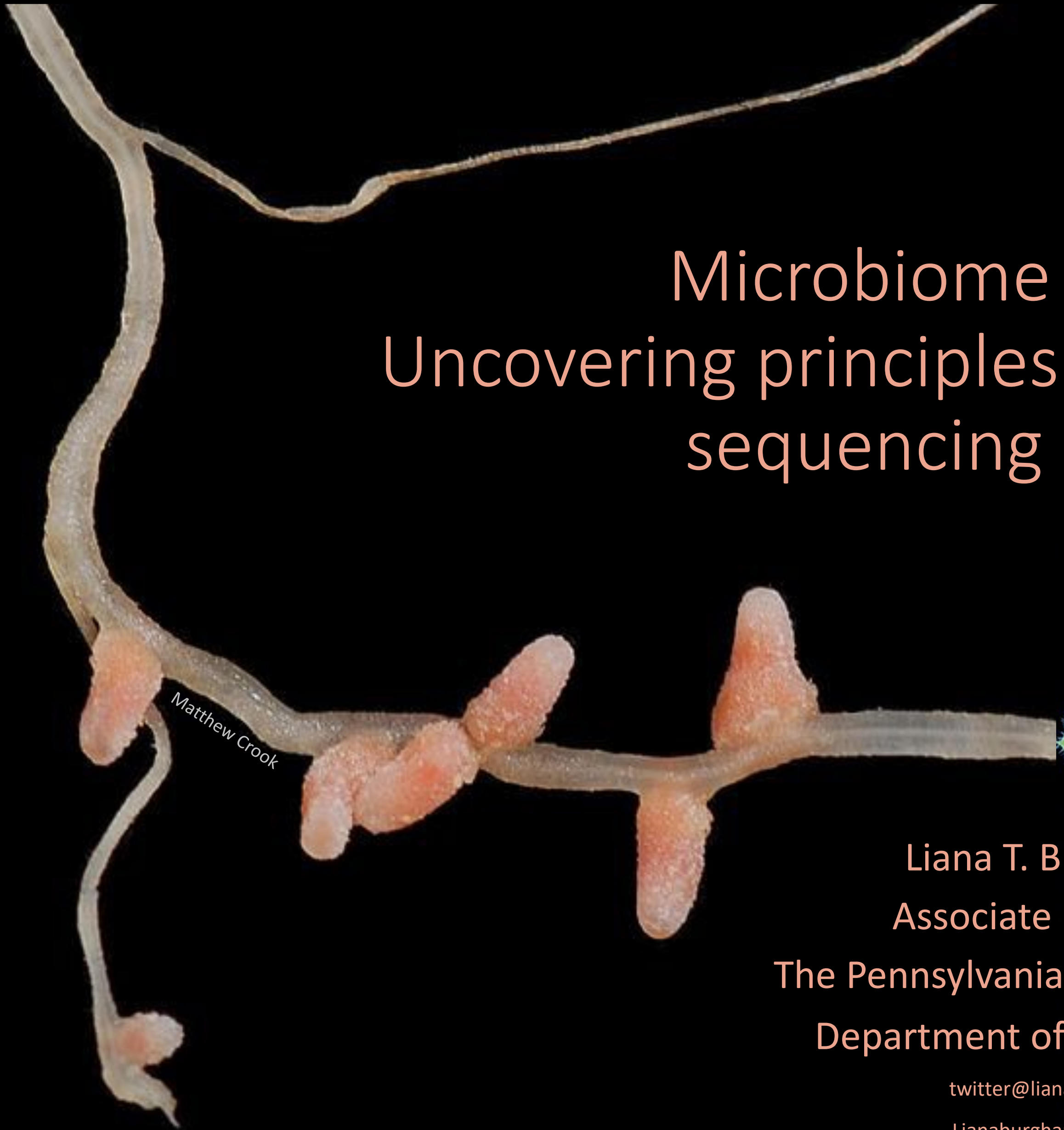
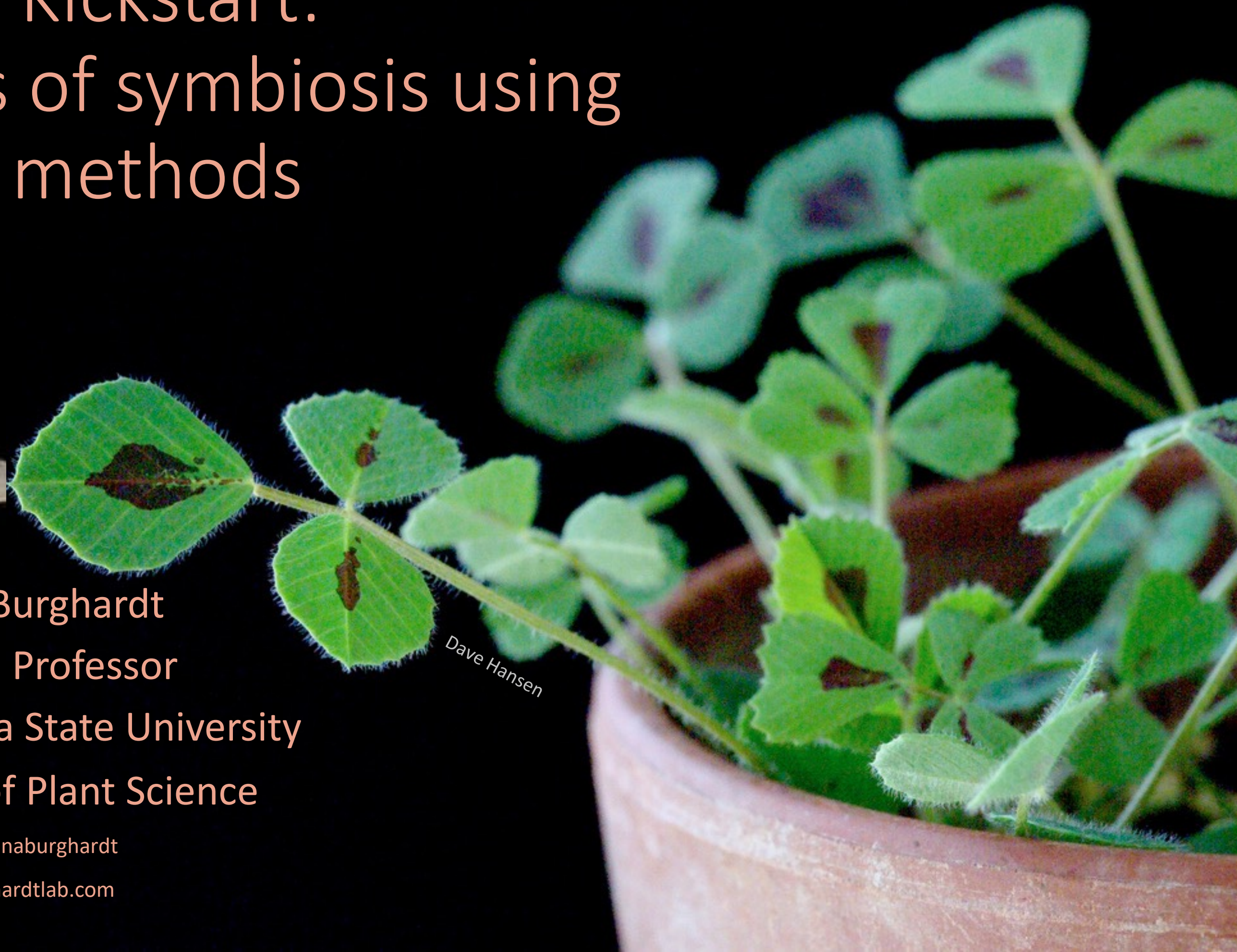
Microbiome Kickstart:
Uncovering principles of symbiosis using sequencing methods





Liana T. Burghardt
Associate Professor
The Pennsylvania State University
Department of Plant Science
twitter@lianaburghardt
Lianaburghardtlab.com

The Burghardt Lab: Examining plant-microbe-climate interactions in natural and managed landscapes

Jennifer Harris
USDA predoc
Ecology Program
Co-adv. E. Couradeau

Kelsey Mercurio
NSF Predoc Fellow
Ecology Program

Patrick Sydow
AEPS Program
SNIP Root Cohort
Co-adv. R. Sawers

Maria Alejandra
Gil Polo
Plant Biology

2020 – present
Asst. Professor

PennState

Amanda Jason
AEPS Program

Cody DePew
Lab Manager

Dr. Sohini Guha
Postdoc

Dr. Kayla Clouse
Postdoc

Learning objectives

- Part 1: Setting the stage for symbiosis
  - Define symbiosis and understand variation in usage
  - Understand basic categories of symbiosis and brainstorm examples
  - Introduce pairwise and holobiont/hologenome frameworks

- Part 2: Leveraging sequencing to study symbiosis
  - Articulate how to use shotgun sequencing to measure symbiont fitness at multiple scales of genetic variation-
  - Fall in love with legumes and rhizobia

What is Symbiosis?

What is Symbiosis?

from Greek συμβίωσις, *symbíōsis*, "living together", from σύν, *sýn*, "together", and βίωσις, bíōsis, "living")

“any type of a close and long-term biological interaction between two biological organisms of different species, termed symbionts, be it mutualism, commensalism, or parasitism”.- Oxford Dictionary

What is Symbiosis?

from Greek συμβίωσις, *symbíōsis*, "living together", from σύν, *sýn*, "together", and βίωσις, bíōsis, "living")

“any type of a close and long-term biological interaction between two biological organisms of different species, termed symbionts, be it mutualism, commensalism, or parasitism”.- Oxford Dictionary

Broad-

"the living together of unlike organisms" 1879, Heinrich Anton de Bary

Heinrich Anton de Bary was a German surgeon, botanist, microbiologist, and mycologist . He is considered a founding father of plant pathology (phytopathology) as well as the founder of modern mycology. His extensive and careful studies of the life history of fungi and

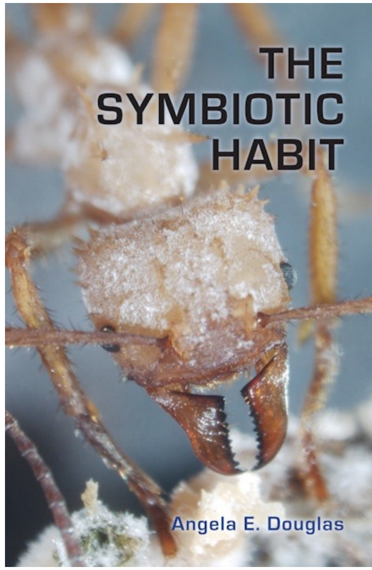
Lynn Margulis was an American evolutionary biologist, and was the primary modern proponent for the significance of symbiosis in evolution. Historian Jan Sapp has said that "Lynn Margulis's name is as synonymous with symbiosis as Charles Darwin's is with evolution." In particular, Margu

Narrow-

“ symbiosis as a persistent mutualism”

Angela Douglas is a proponent of this framing.

Used colloquially and by general biologists



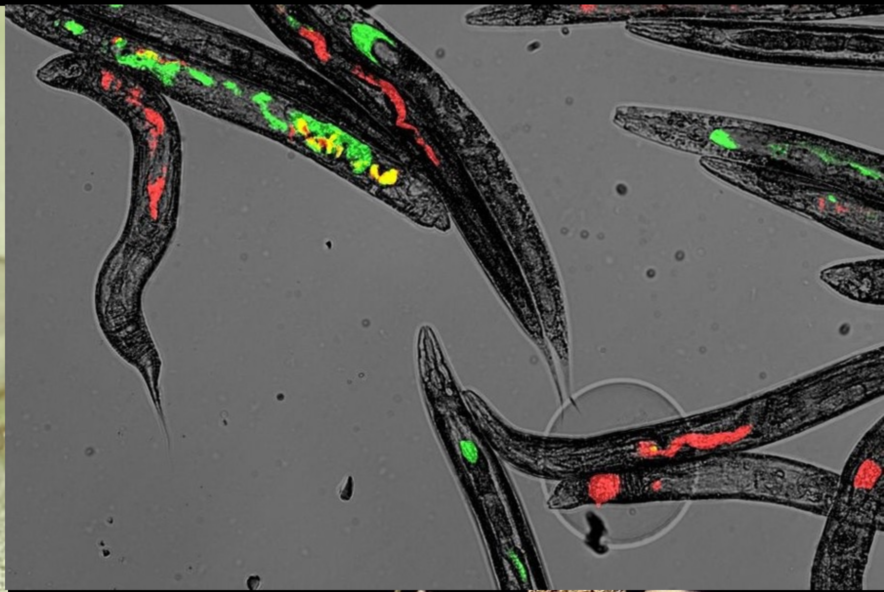
A non-exclusive list of axis of symbiosis variation!

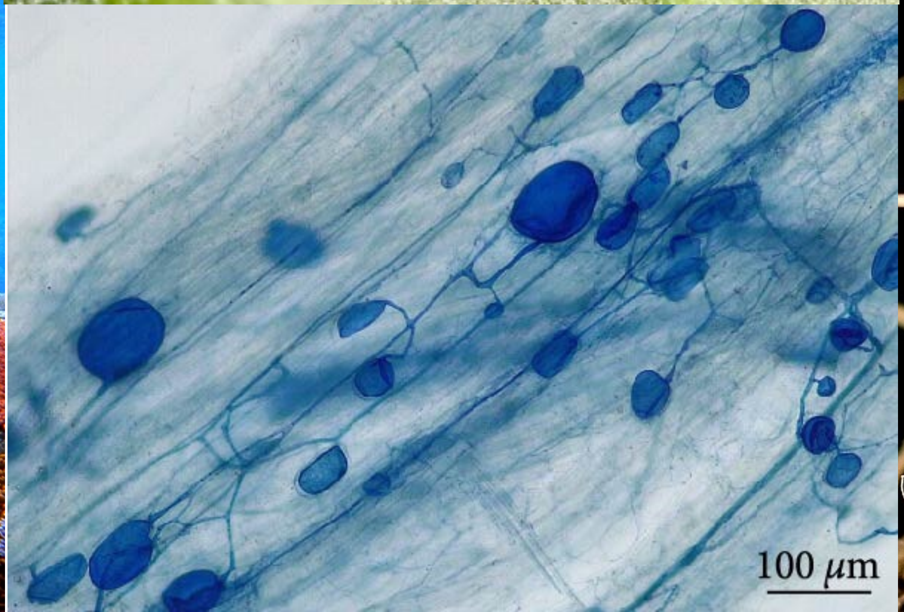
- Ecological outcome—mutualism, commensalism, parasitism
- Dependence—obligate vs. facultative
- Transmission—environmental vs. horizontal vs. vertical
- Participants—one-to-one vs. many-to-one
- Location– ecto vs. endo
- Generational—synchronous vs. asymmetric
- Others?

Brainstorm two examples of a symbiosis (one has to involve a microbe)!

Brainstorm two examples of a symbiosis (one has to have a microbe)!







Think/Pair/Share:
See if you can slot your examples into the following categories

- Ecological outcome—mutualism, commensalism, parasitism
- Dependence—obligate vs. facultative
- Transmission—horizontal vs. vertical
- Participants—one-to-one vs. many-to-one
- Location– ecto vs. endo
- Generational—synchronous vs. asymmetric
- Others?

Why should microbiome scientists care about the symbiosis concept?

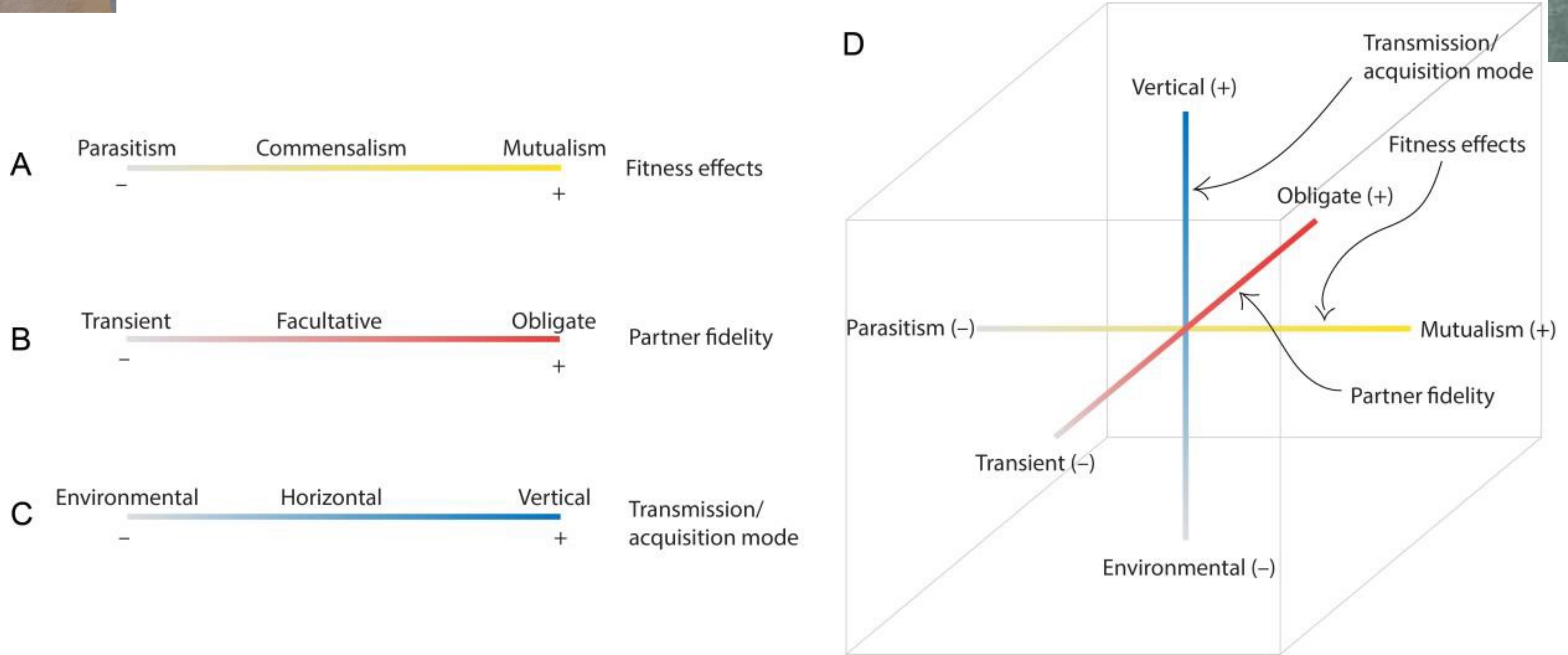
Why should microbiome scientists care about the symbiosis concept?

Lily Khadempour

EcoEvoRxiv:
A new conceptual framework for host-microbe symbiosis

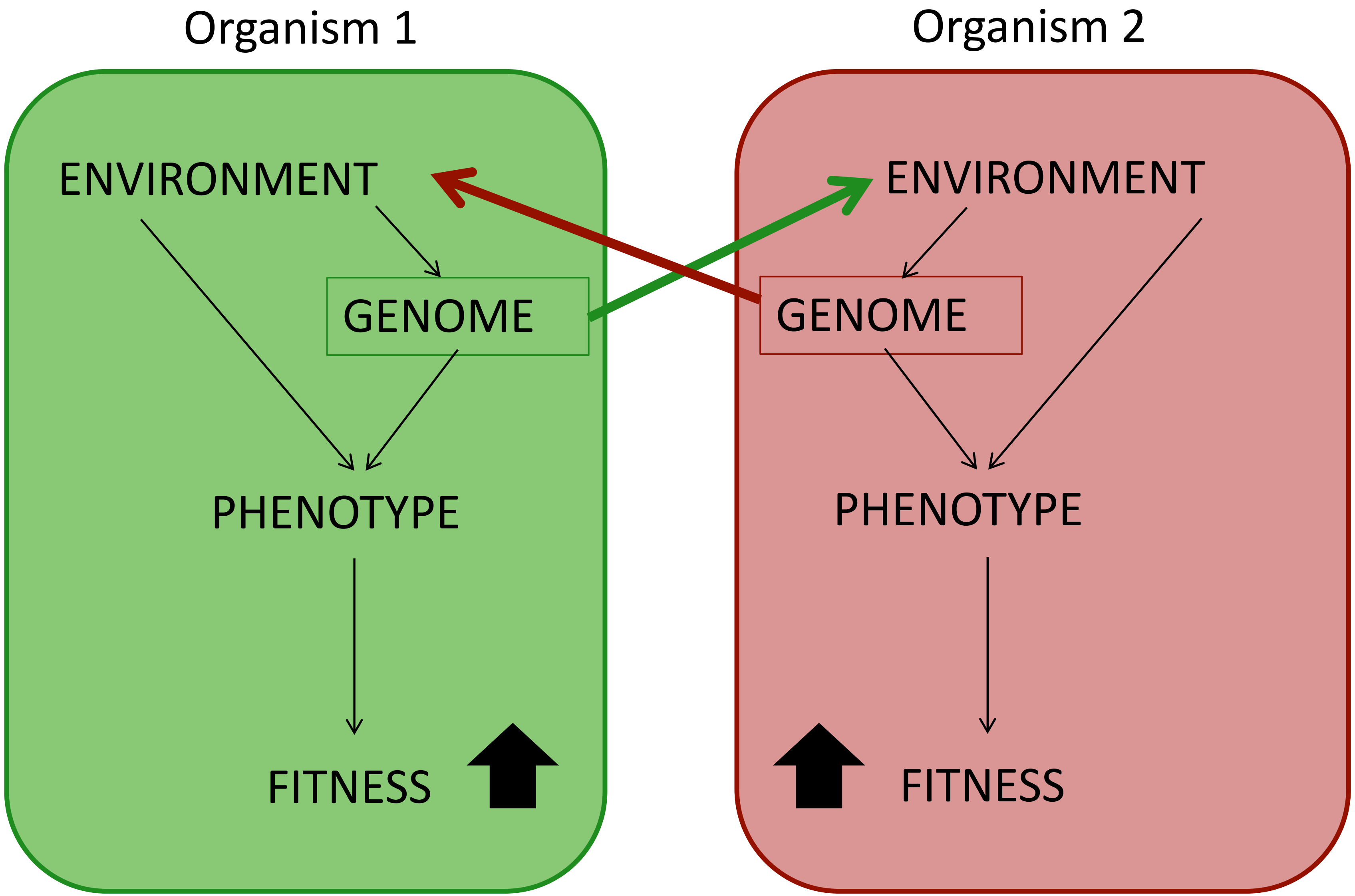
mGem:
Tapping into the language of symbiosis to advance human microbiome research

Gina Lewin



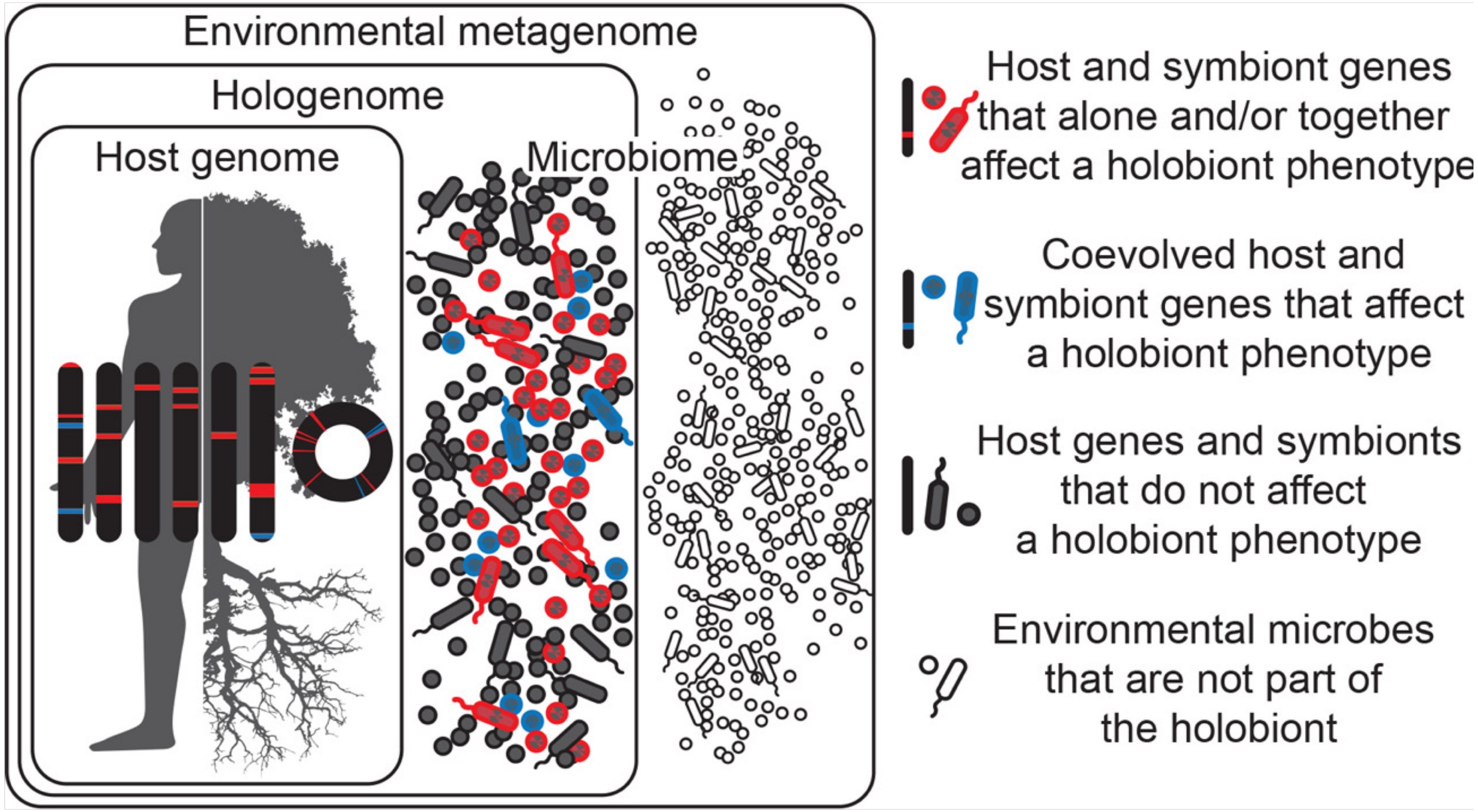
Why should microbiome scientists care about the symbiosis concept?

One to one genome interaction model…



Interacting genomes in symbiosis

- **Holobiont** is a term used to describe an individual host and its microbial community, including viruses and cellular microorganisms
- It is derived from the Greek word holos, which means whole or entire.
- Lynn Margulis was the first to coin the term
- **Hologenome**-encompass the genomes of the host and all of its microbes at any given time point
- **Hologenome evolutionary theory** (still being sorted out…)



Theis… Bordenstein et. al , 2016

Learning objectives

- Part 1: Setting the stage for symbiosis
  - Define symbiosis and understand variation in usage
  - Understand basic categories of symbiosis and brainstorm examples
  - Introduce pairwise vs. holobiont/hologenome frameworks
- Part 2: Leveraging sequencing to study symbiosis
  - Articulate how to use shotgun sequencing to measure symbiont fitness at multiple scales of genetic variation-
  - Fall in love with legumes and rhizobia

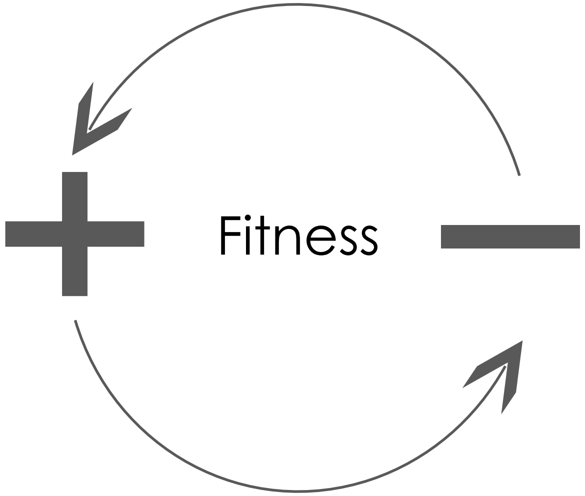
Learning objectives

- Part 1: Setting the stage for symbiosis
  - Define symbiosis and understand variation in usage
  - Understand basic categories of symbiosis and brainstorm examples
  - Introduce pairwise vs. holobiont/hologenome frameworks
- Part 2: Leveraging sequencing to study symbiosis
  - Articulate how to use shotgun sequencing to measure symbiont fitness at multiple scales of genetic variation-
  - Fall in love with legumes and rhizobia

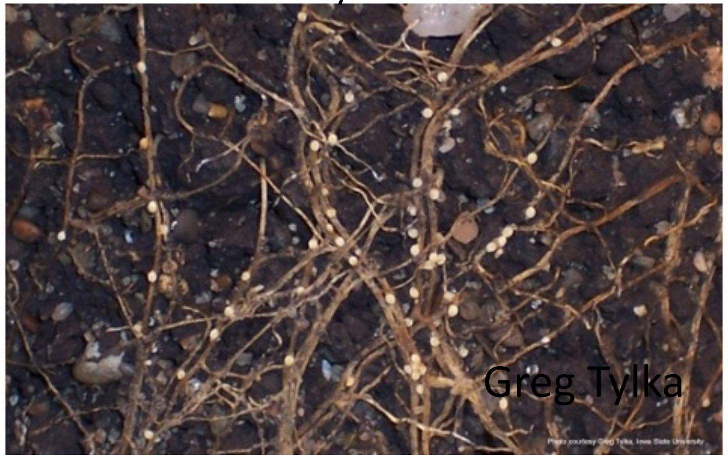
Symbiosis influences host growth & nutrition

Soybean cyst nematode

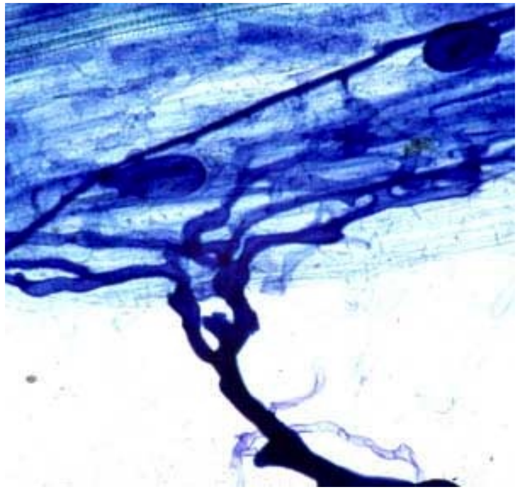
Parasitism



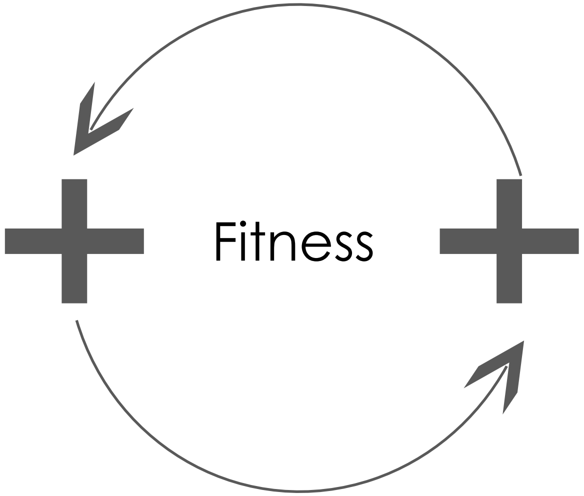
Soybean



Arbuscular mycorrhizal fungi



Mutualism

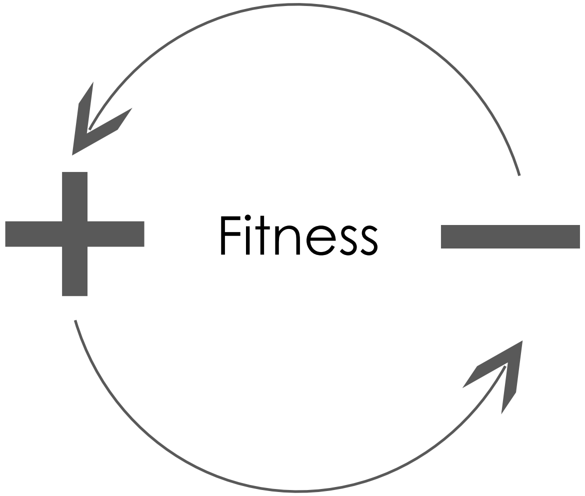


Corn

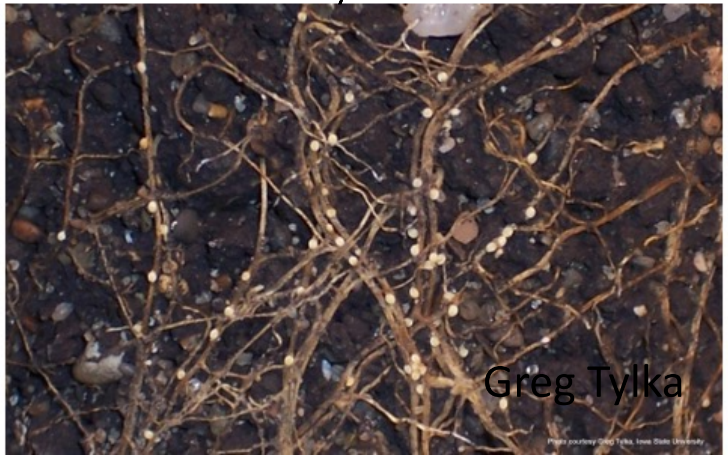
It is rare to measure the relative fitness of microbial symbionts

Soybean cyst nematode

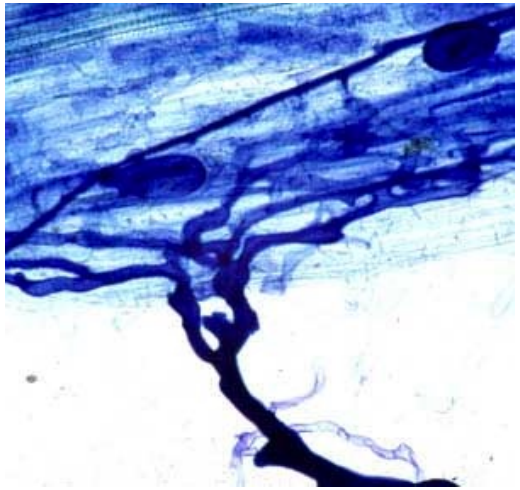
Parasitism



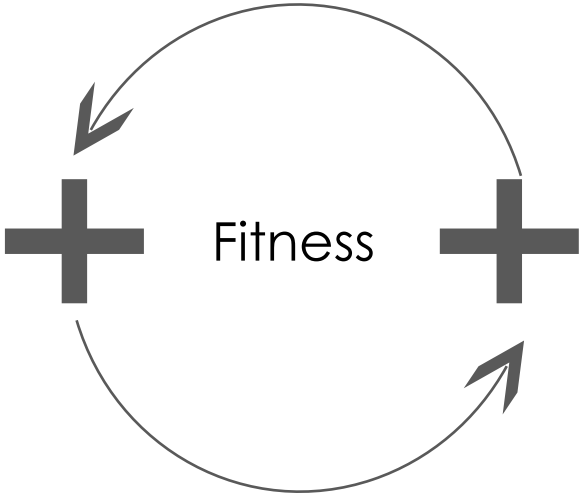
Soybean



Arbuscular mycorrhizal fungi

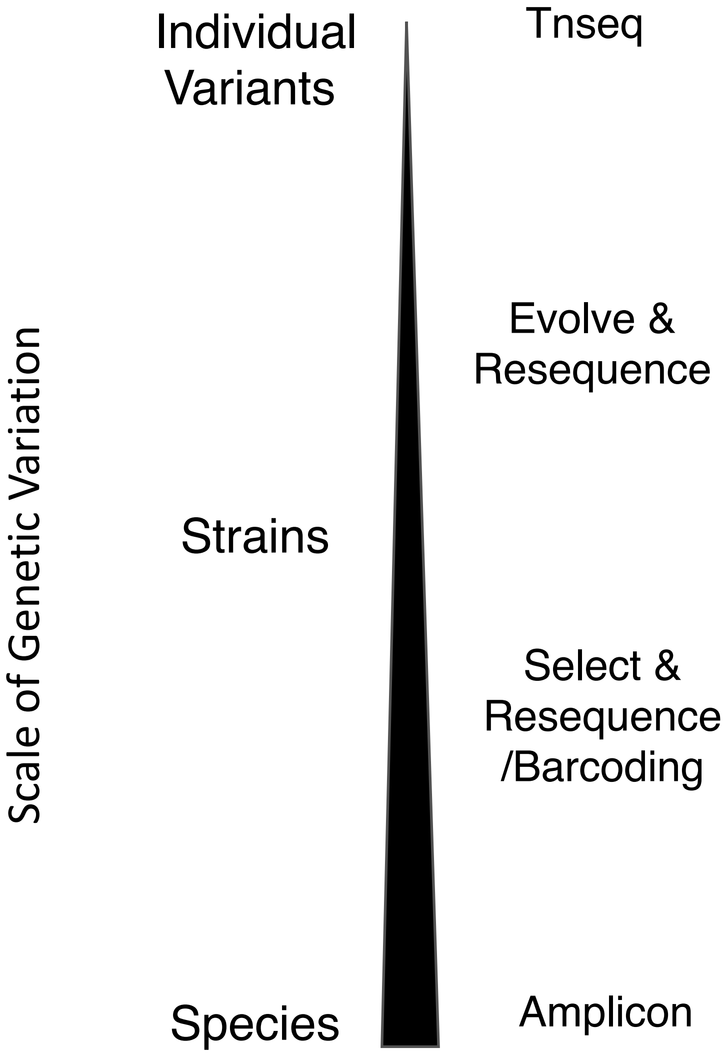


Mutualism

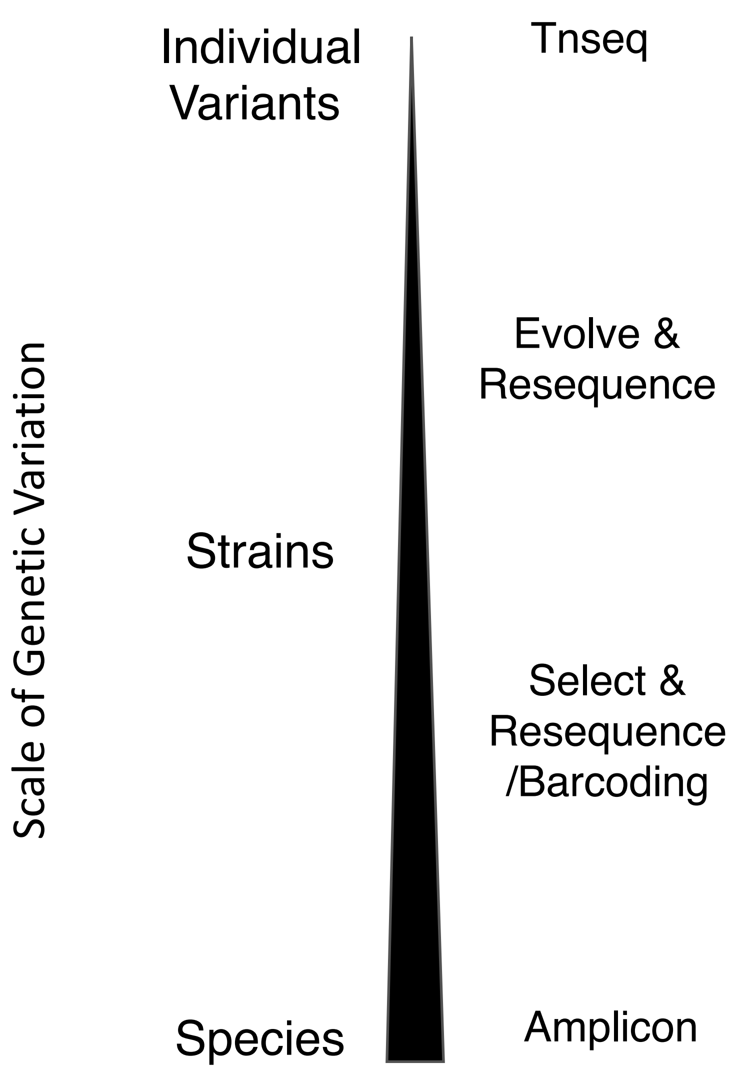


Corn

Whole genome shotgun sequencing allows measurement of microbial symbiont fitness at many different levels of genetic variation



Whole genome shotgun sequencing allows measurement of microbial symbiont fitness at many different levels of genetic variation



PNAS

Lifestyle adaptations of *Rhizobium* from rhizosphere to symbiosis

Rachel M. Wheatley[a,1], Brandon L. Ford[a,1], Li Li[b,1], Samuel T. N. Aroney[a], Hayley E. Knights[a], Raphael Ledermann[a], Alison K. East[a], Vinoy K. Ramachandran[a,2], and Philip S. Poole[a,2]

ADAPTATION

Science

Experimental evolution makes microbes more cooperative with their local host genotype

Rebecca T. Batstone[1,2*], Anna M. O'Brien[1,3], Tia L. Harrison[1], Megan E. Frederickson[1,4,5]

PNAS

Select and resequence reveals relative fitness of bacteria in symbiotic and free-living environments

Liana T. Burghardt[a], Brendan Epstein[a], Joseph Guhlin[a], Matt S. Nelson[b], Margaret R. Taylor[b], Nevin D. Young[a,c], Michael J. Sadowsky[a,b,d], and Peter Tiffin[a,1]

ISME

Variation in rhizosphere microbial communities and its association with the symbiotic efficiency of rhizobia in soybean

Qin Han[1] · Qun Ma[1] · Yong Chen[1] · Bing Tian[1] · Lanxi Xu[1] · Yang Bai[2] · Wenfeng Chen[3] · Xia Li[1]

Legumes & Rhizobia

Trifolium pretense

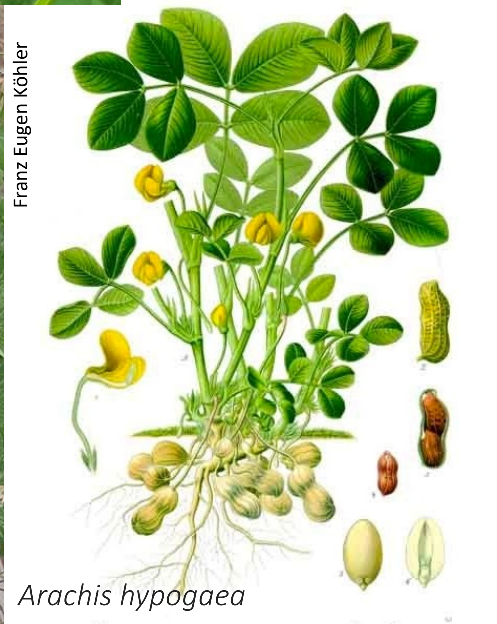

Glycine max

Acacia cornigera

Hippophae rhamnoides L.

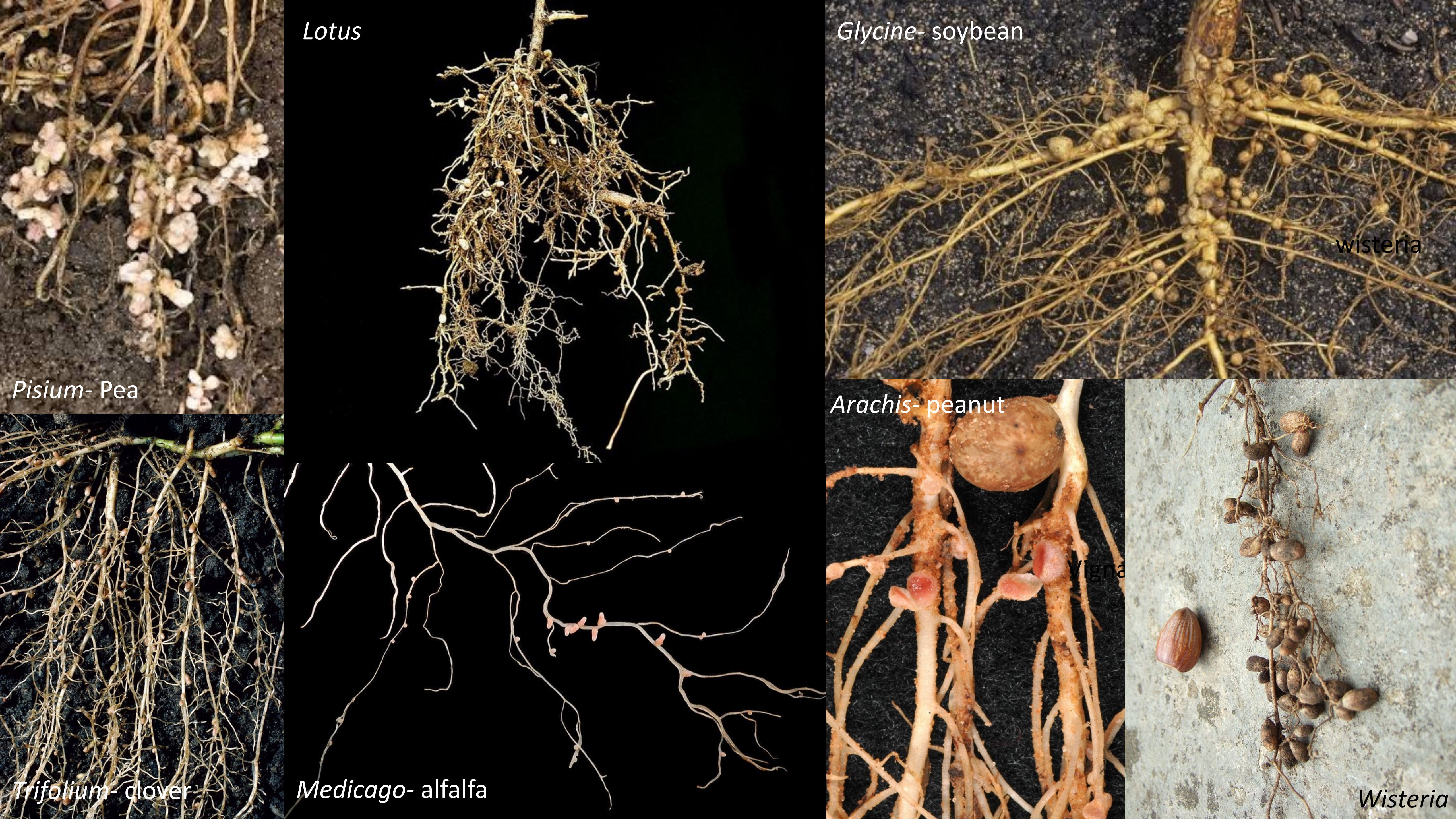

Lens culirnaris

Pisum sativum



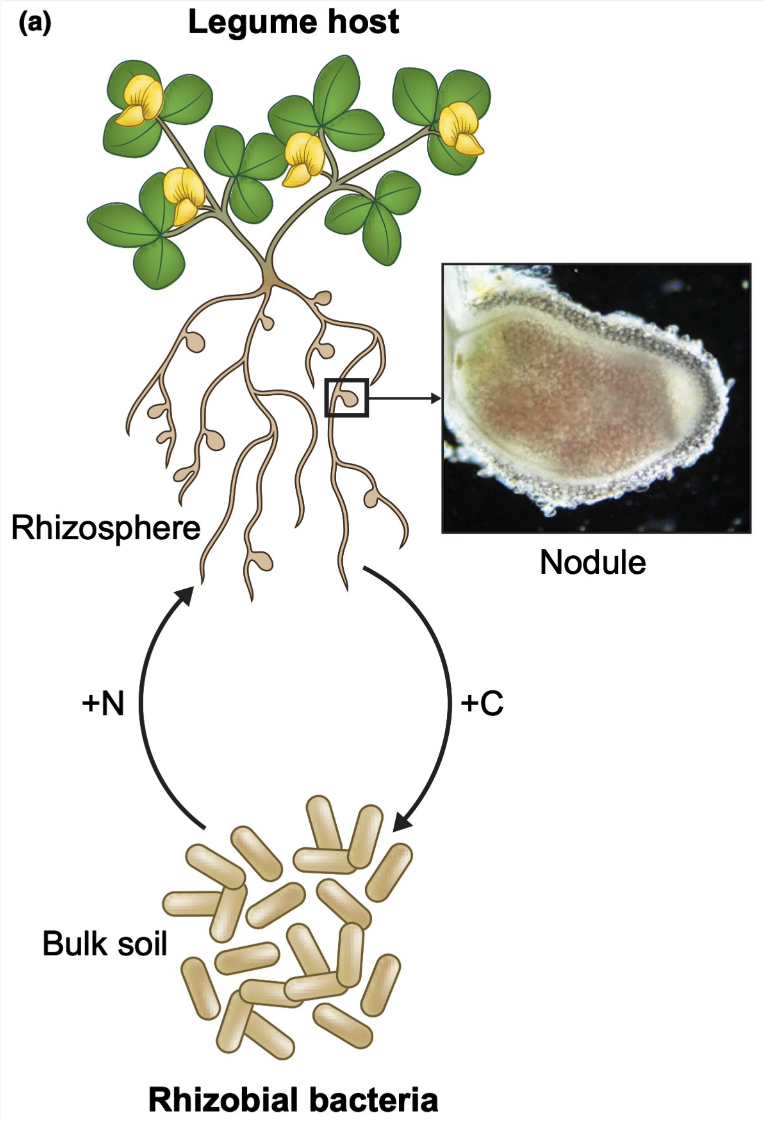
Arachis hypogaea


Lupinus perennis



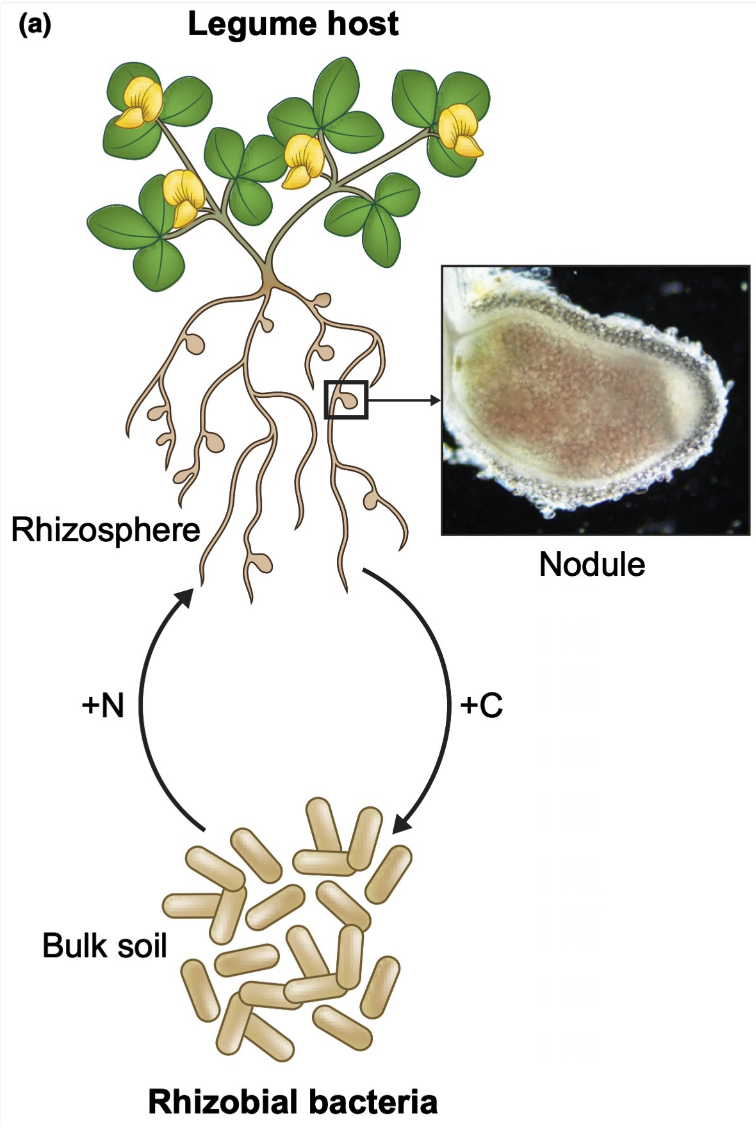
Pisium- Pea
Lotus
Glycine- soybean
wisteria
Arachis- peanut
Trifolium- clover
Medicago- alfalfa
Wisteria

Facultative symbiosis are ubiquitous, important, and poorly understood from the microbial perspective

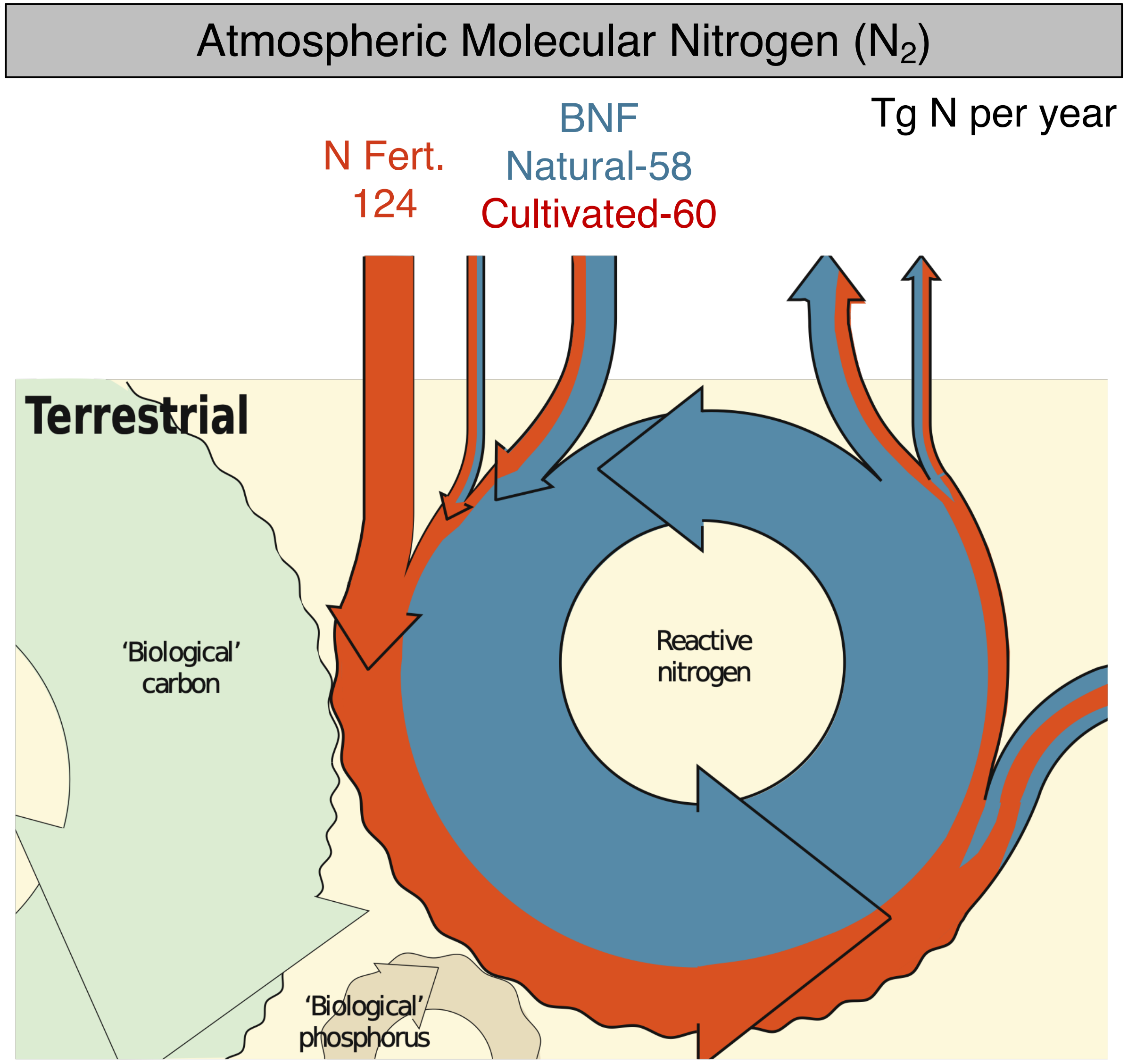


(Burghardt et al. 2020, *New Phytologist*)

Facultative symbiosis are ubiquitous, important, and poorly understood from the microbial perspective

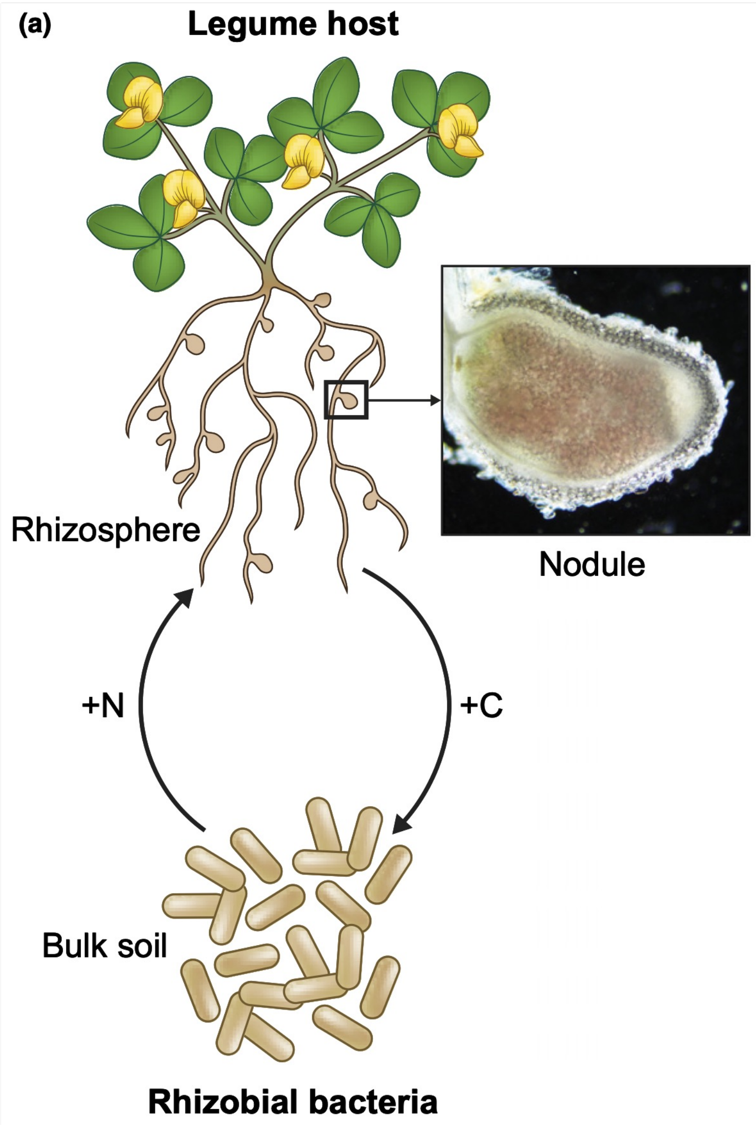


(Burghardt et al. 2020, *New Phytologist*)

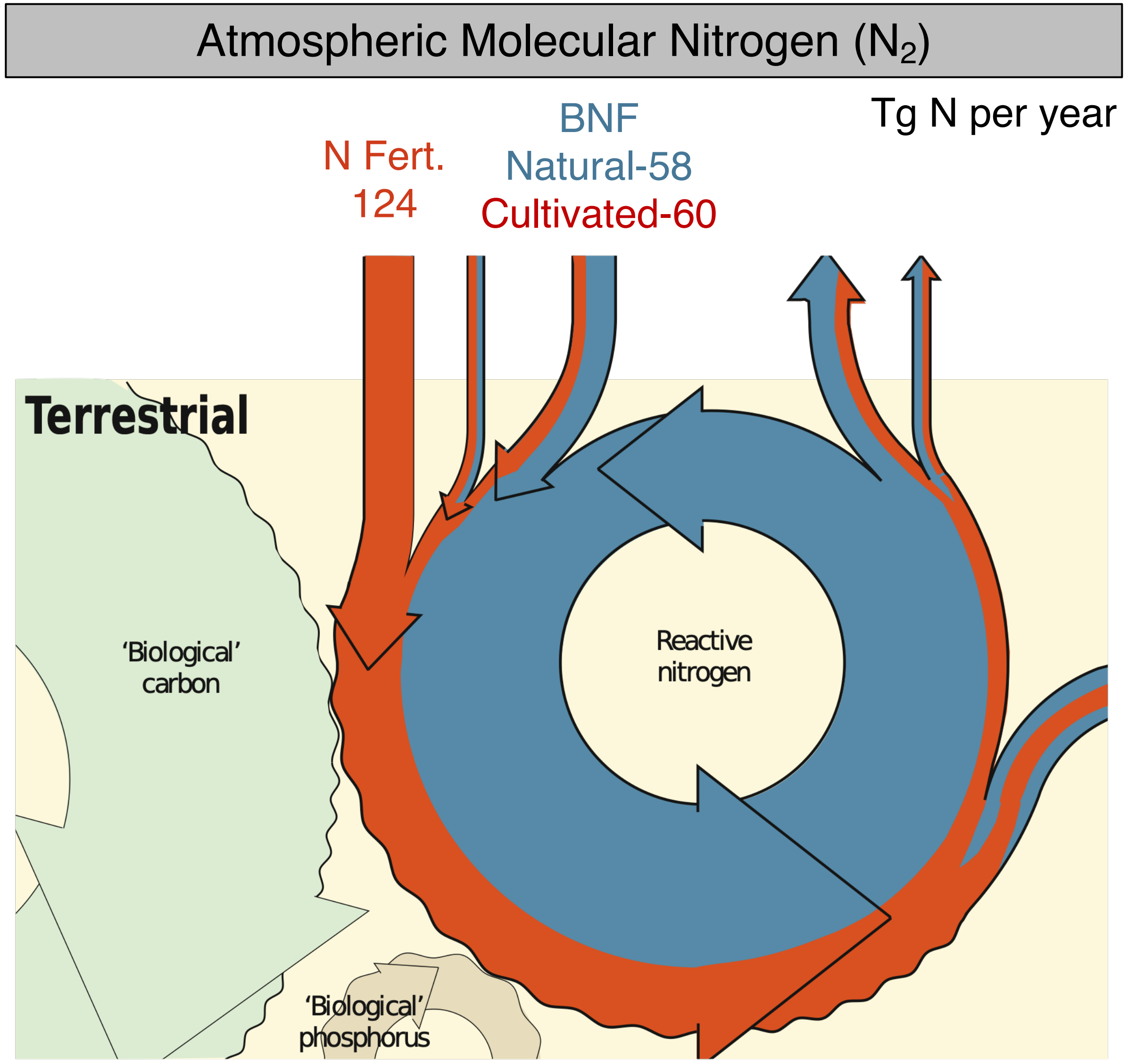


Modified from: Gruber & Galloway 2008 with updated fluxes from IPCC Chapter 6 2013

Facultative symbiosis are ubiquitous, important, and poorly understood from the microbial perspective



(Burghardt et al. 2020, *New Phytologist*)



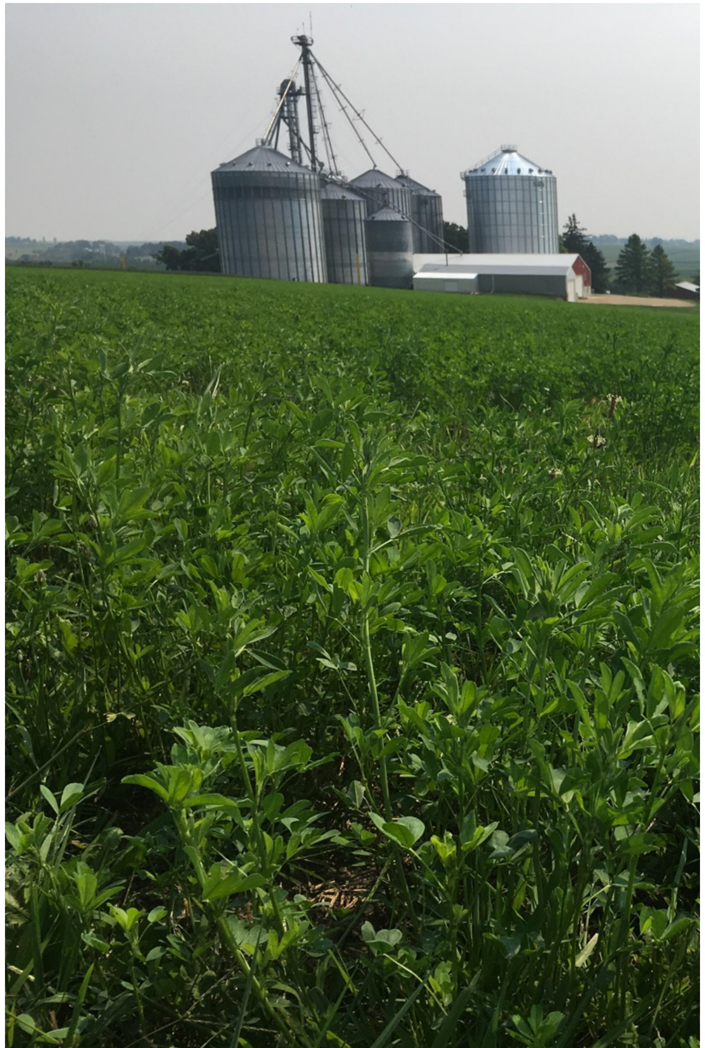
Modified from: Gruber & Galloway 2008 with updated fluxes from IPCC Chapter 6 2013

Barriers to managing Biological Nitrogen Fixation

- Failure to establish
- Competition problem
- Mutualism Breakdown

Legume-Rhizobia System: *Medicago-Sinorhizobium*

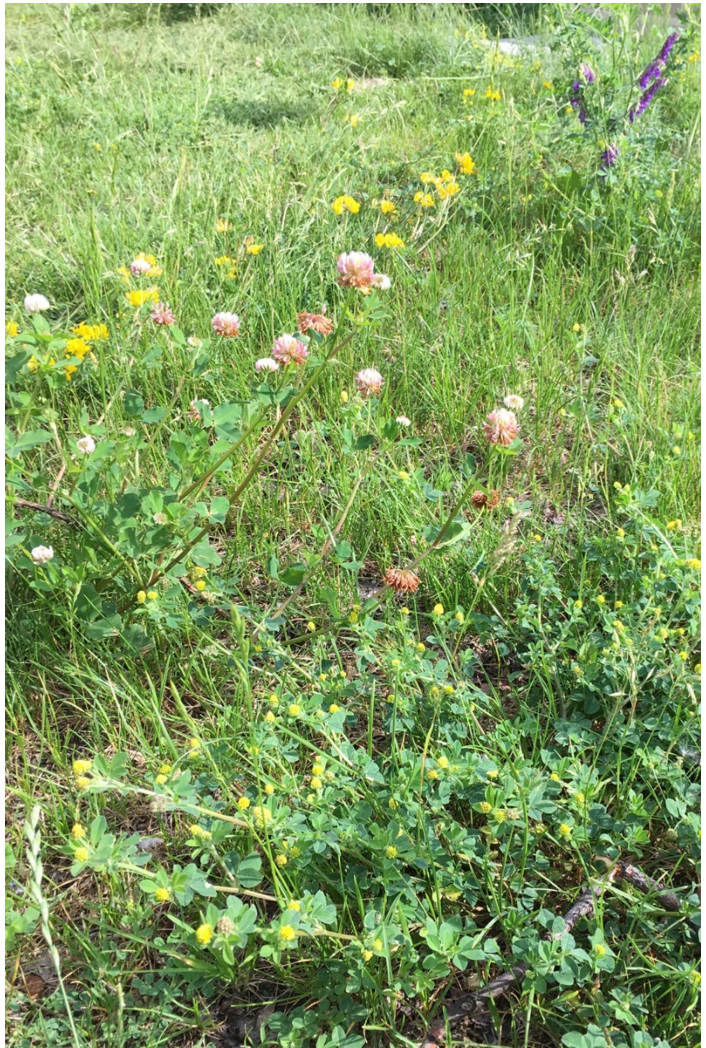
Crop:
M. sativa (alfalfa)



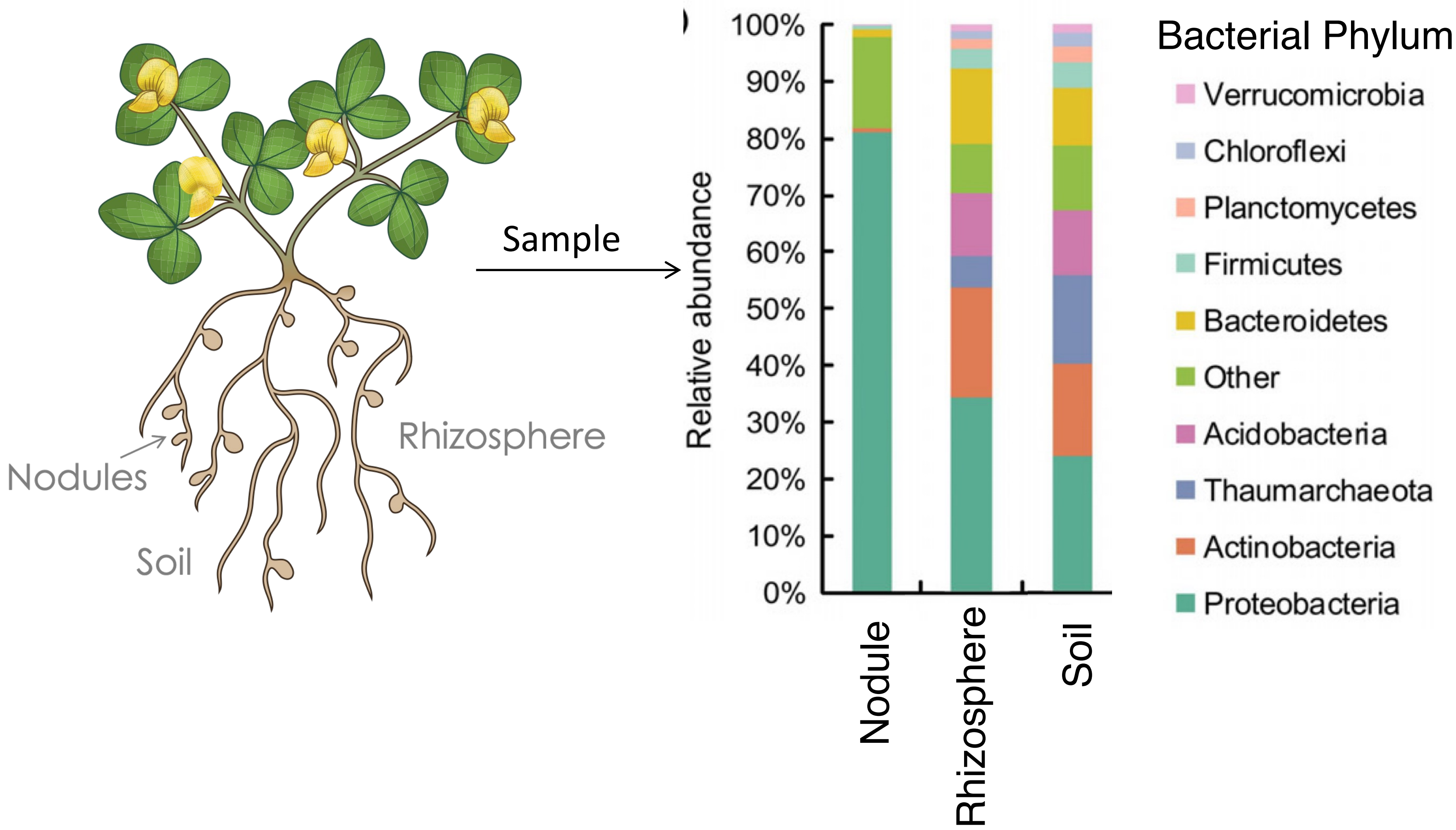
Genetic model:
Medicago truncatula

Genetic model:
Sinorhizobium meliloti

Naturalized species:
M. lupulina; M. polymorpha

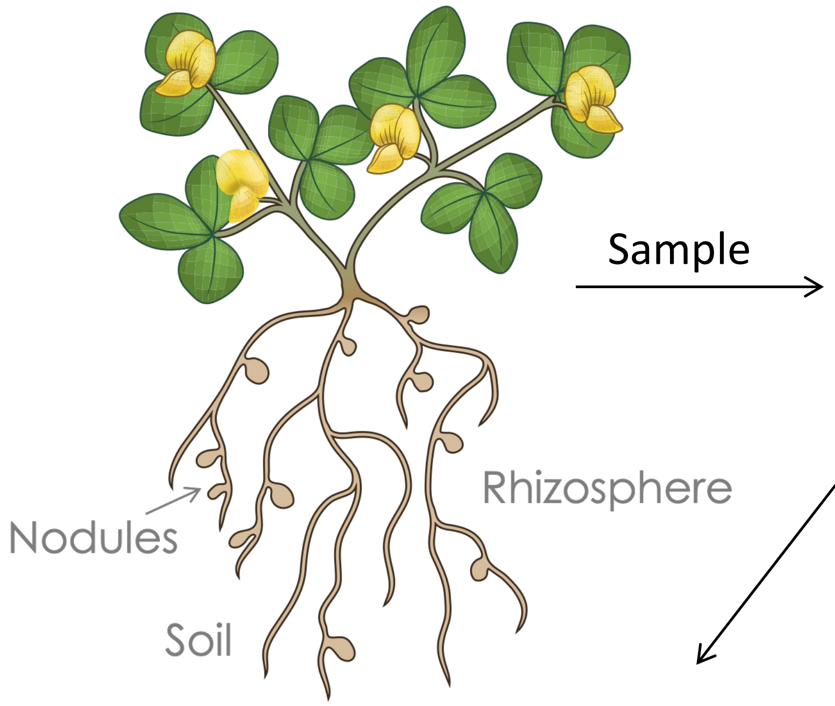


Medicago nodules are highly selective

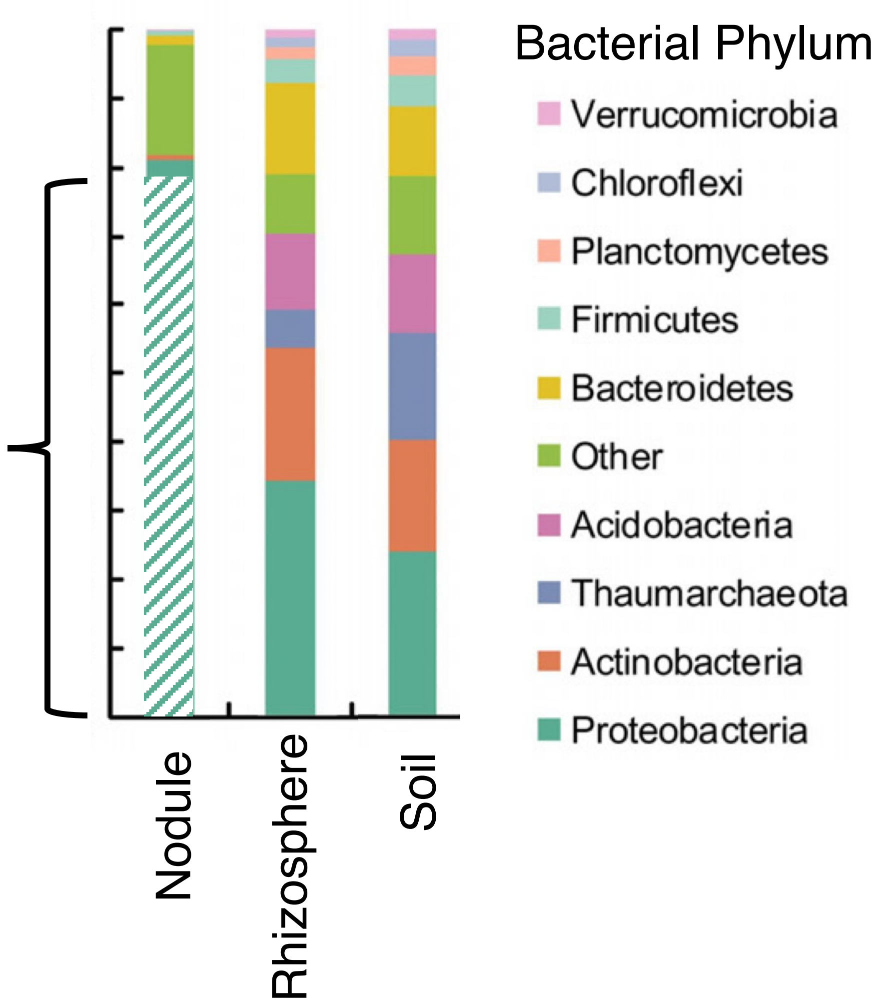


Xiao Xiao et. al. Molecular Ecology 2017

Medicago nodules are highly selective

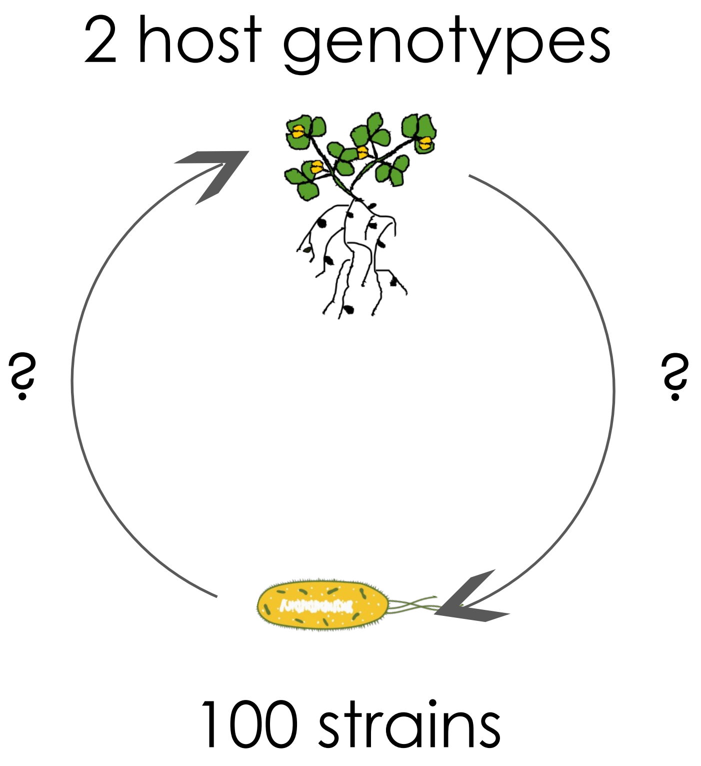


97% Sinorhizobium meliloti & medicae



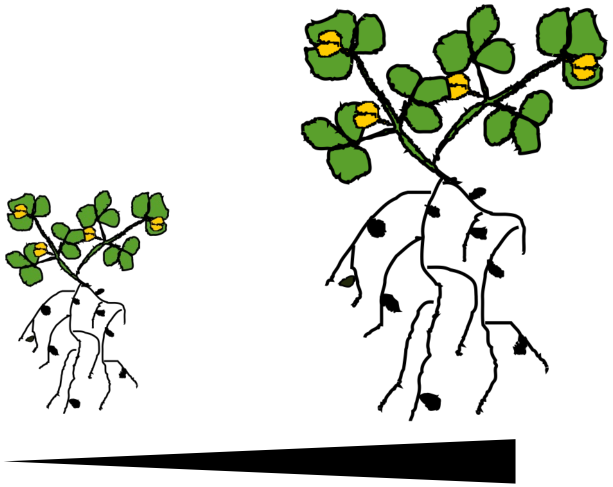
Xiao Xiao et. al. Molecular Ecology 2017

Traditional single-strain experiments to test effect of intra-specific genetic variation on partner fitness



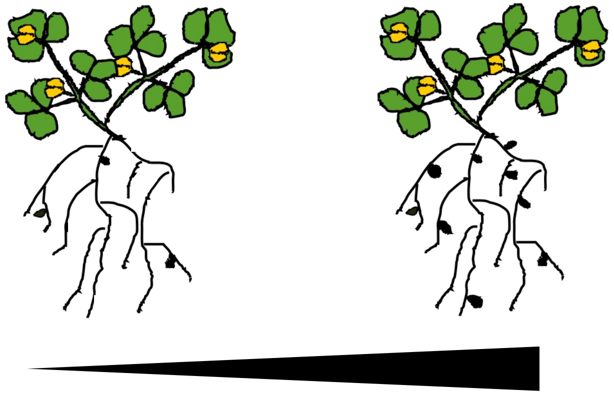
Measure fitness proxies: traits correlated w/ fitness

+ Plant Fitness



Plant biomass

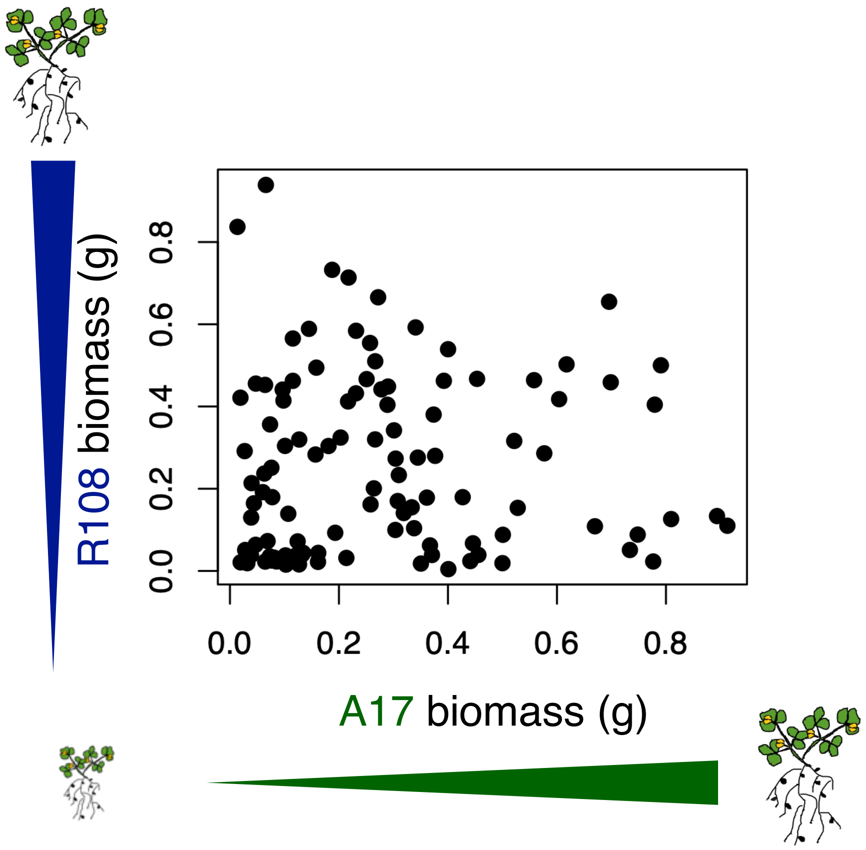
Bacterial Fitness +



Nodule number
Nodule size

Intra-specific strain variation influences plant fitness

Plant Fitness

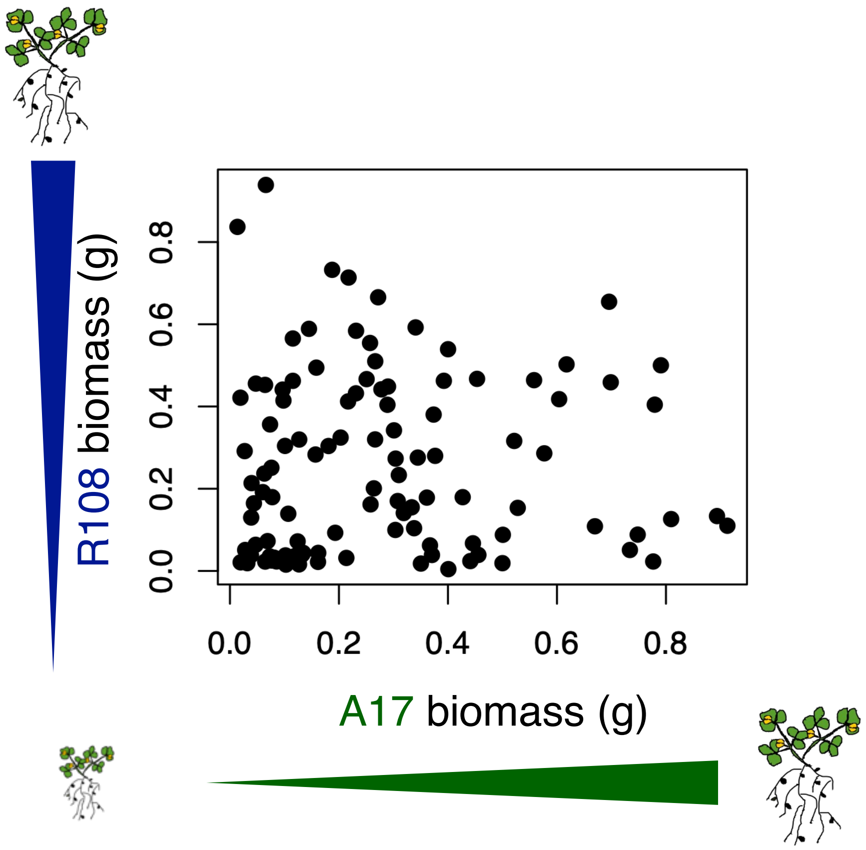


All of this functional variation occurs within a single ASV/bacterial species!!!!!

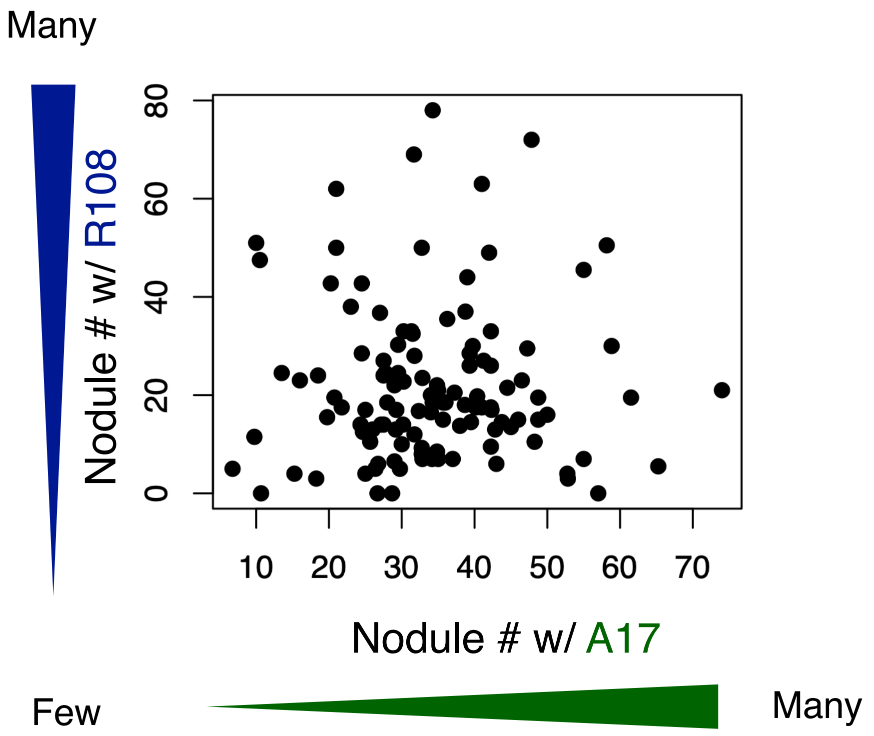
Burghardt et al. 2018, *PNAS*

Intra-specific host variation influences bacterial fitness

Plant Fitness



Bacterial Fitness



Burghardt et al. 2018, *PNAS*

In nature… strains don't exist alone! So how can we measure symbiont fitness?

Burghardt et al. 2018, *PNAS*

A microbial triathlon: Routes to shaping evolutionary trajectories

Nodules

Rhizosphere

Soil

[Culture]

Selection

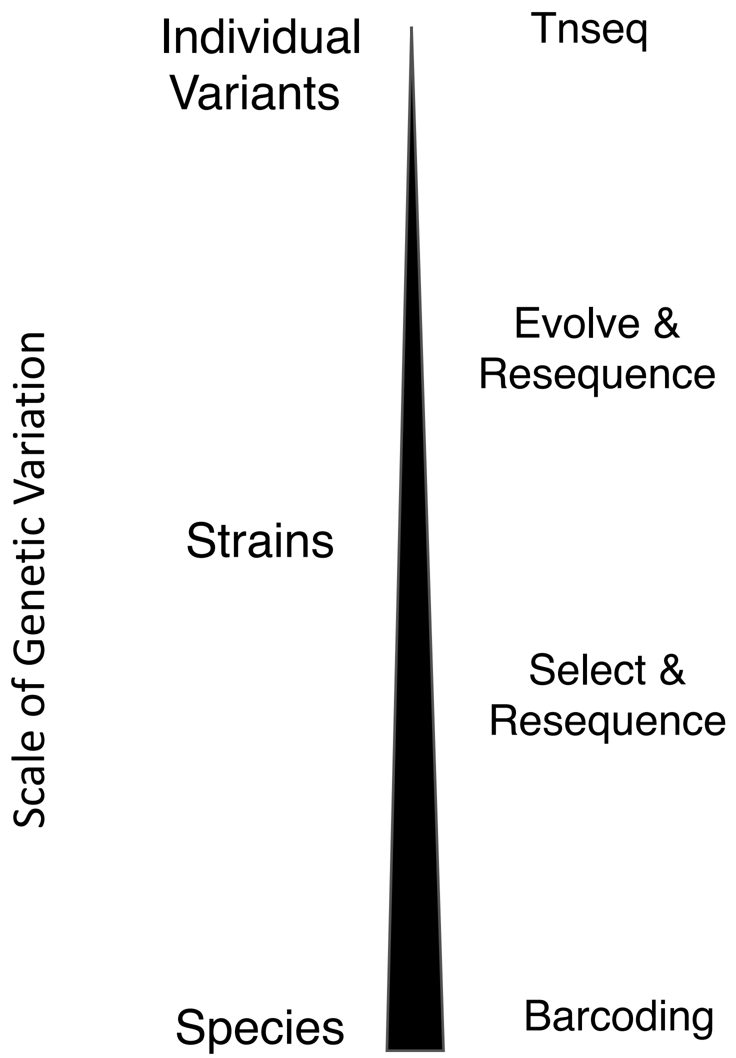
Population size

Frequency of exposure

Drift

Mutation

Whole genome shotgun sequencing allows measurement of microbial symbiont fitness at many different levels of genetic variation

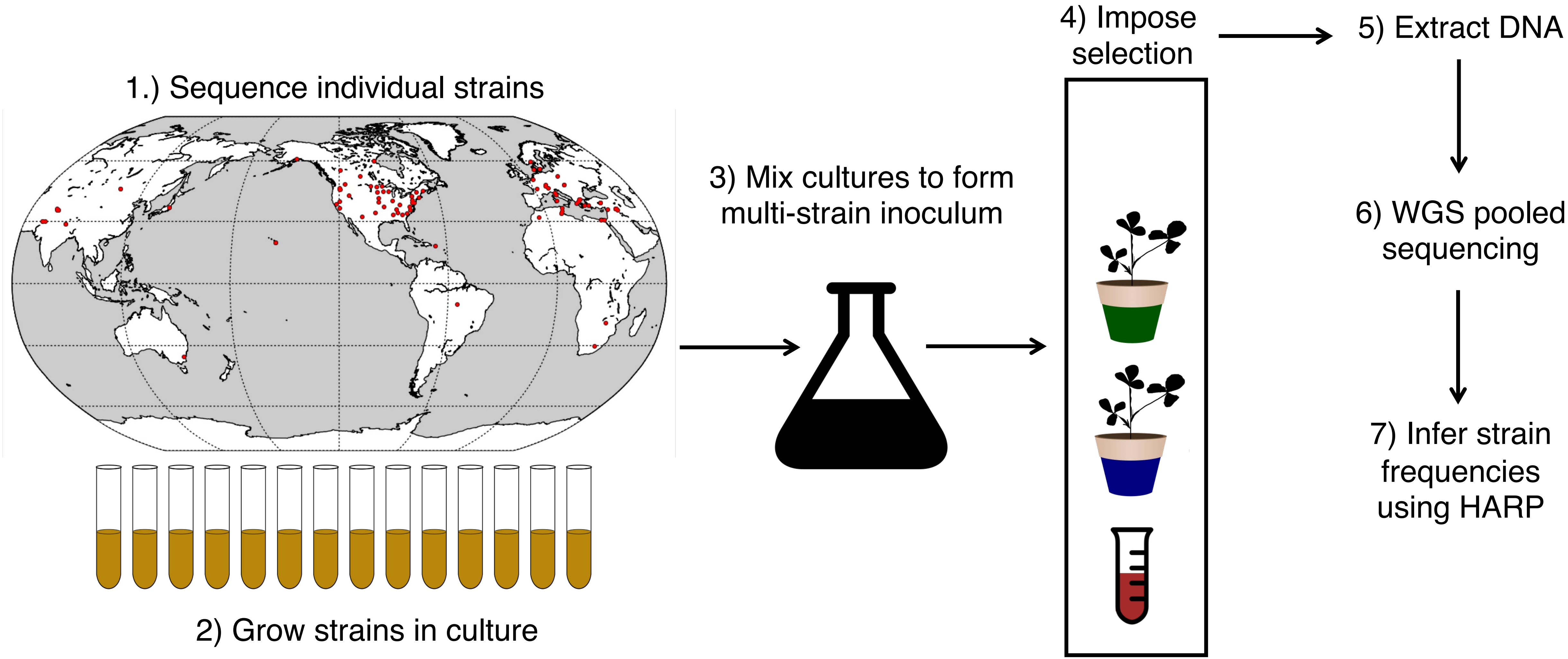


PNAS

Select and resequence reveals relative fitness of bacteria in symbiotic and free-living environments

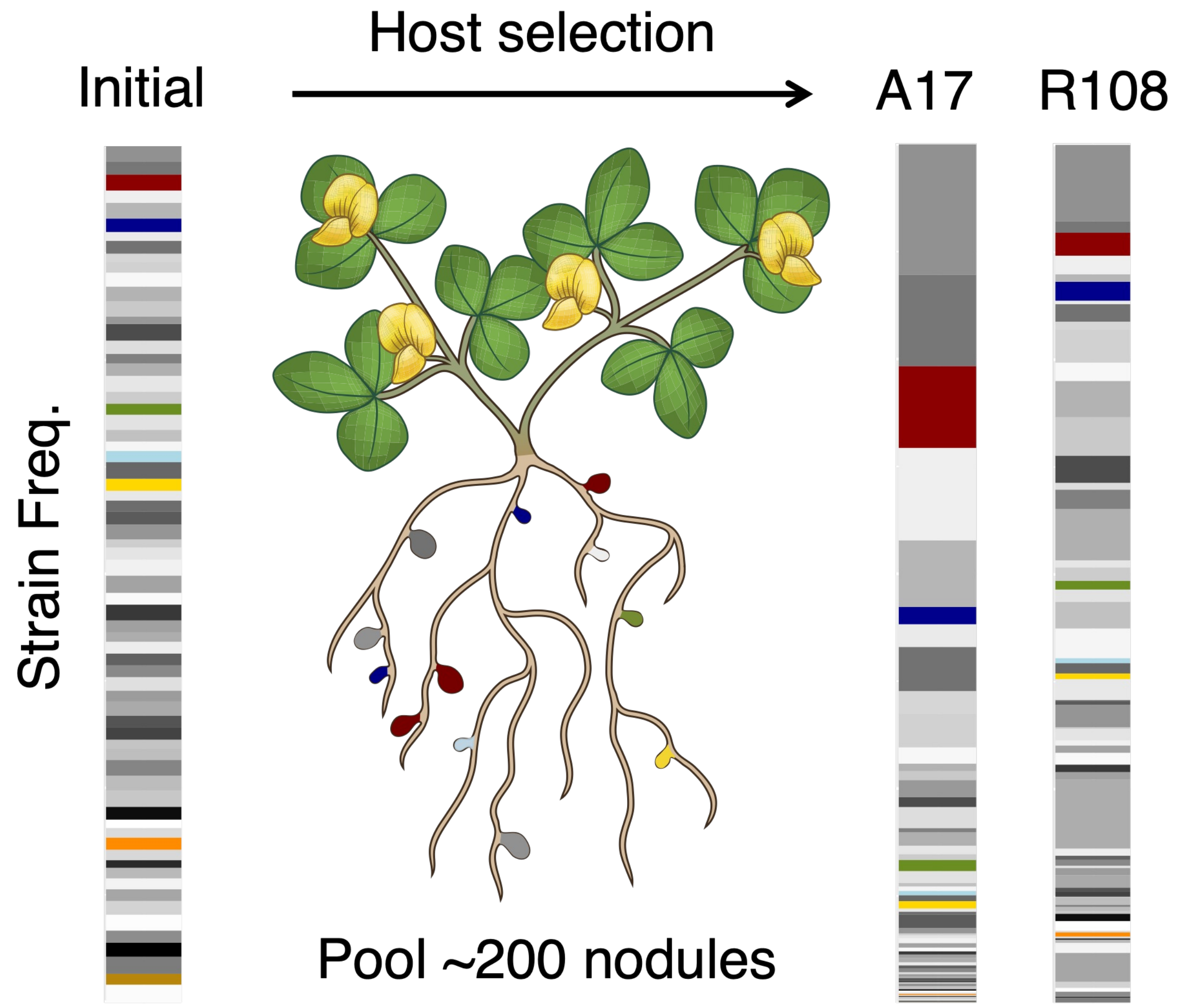
Liana T. Burghardt[a], Brendan Epstein[a], Joseph Guhlin[a], Matt S. Nelson[b], Margaret R. Taylor[b], Nevin D. Young[a,c], Michael J. Sadowsky[a,b,d], and Peter Tiffin[a,1]

‘Select and Resequence’ Method

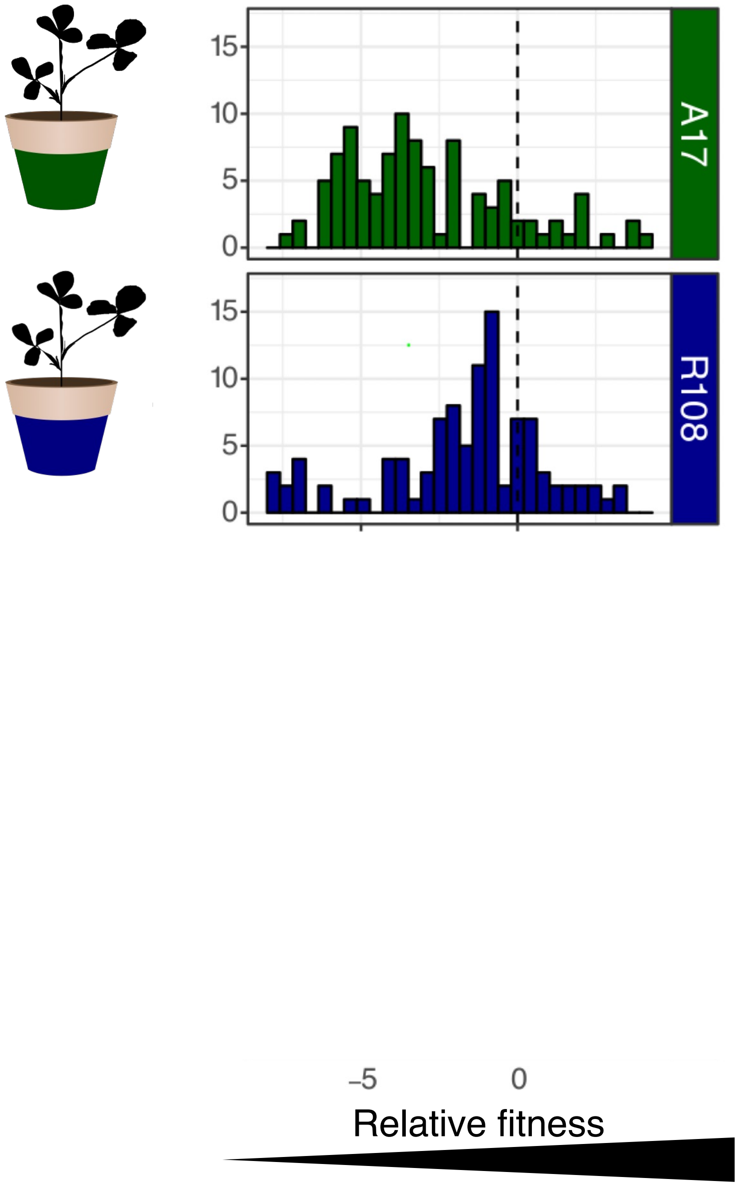


Burghardt et al. 2018, *PNAS*

Medicago genotypes enrich different strains

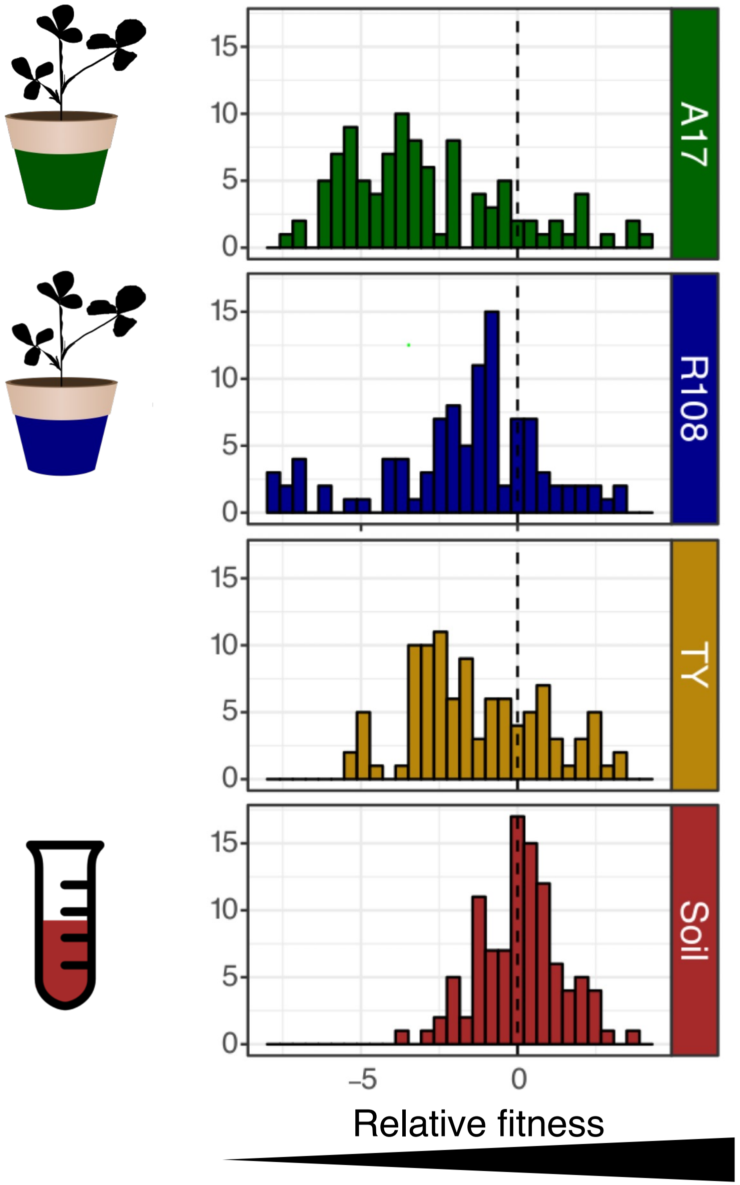


Convert strain frequencies into relative fitness



Burghardt et al. 2018, *PNAS*

Hosts impose strong selection, reducing diversity

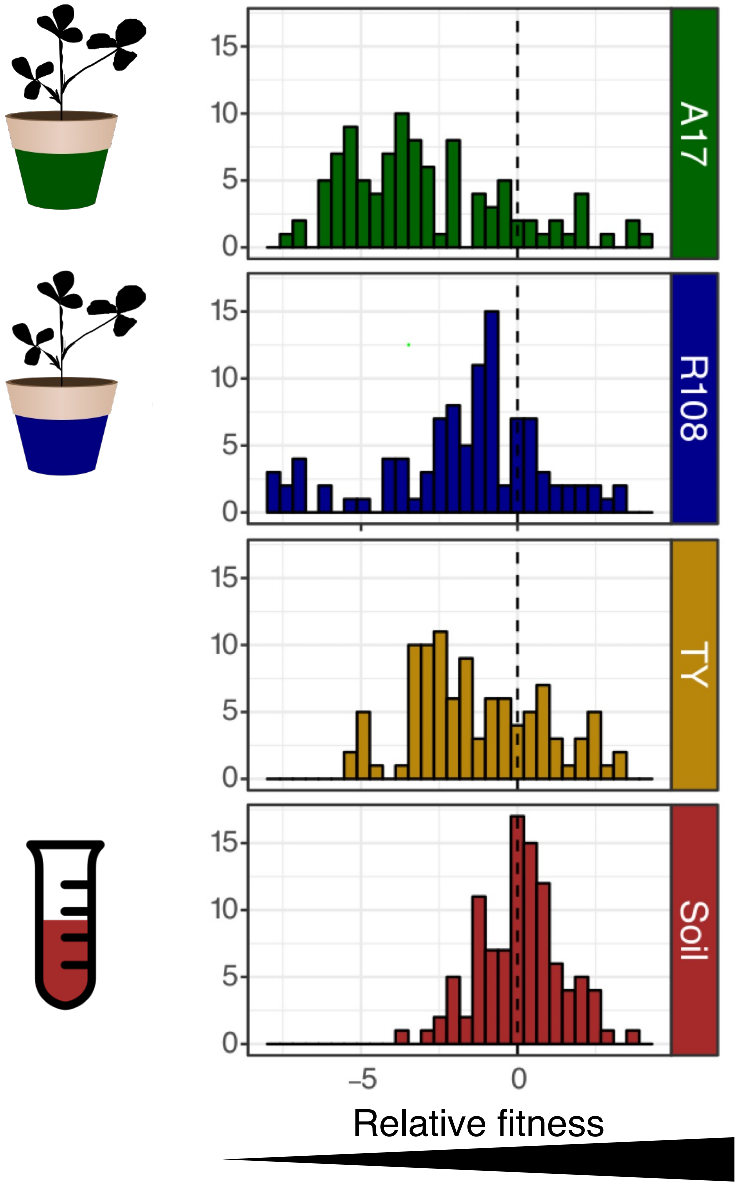


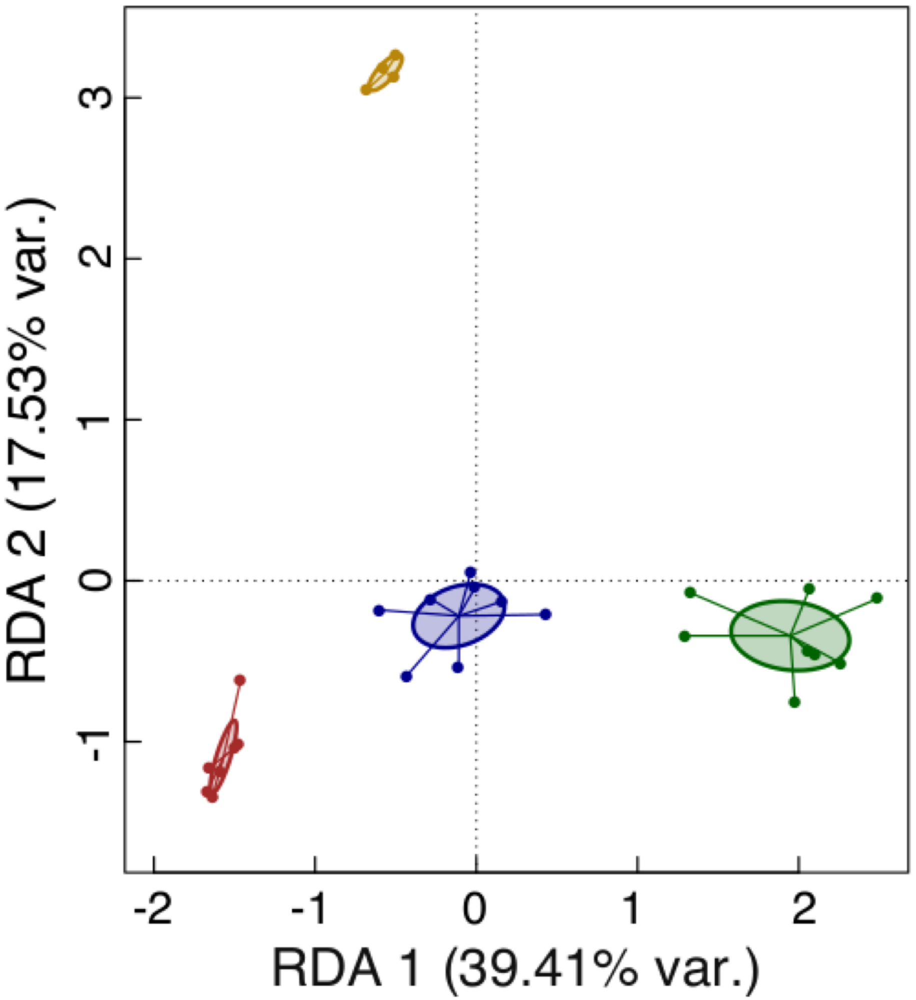
Stronger

Weaker

Burghardt et al. 2018, *PNAS*

Strains exhibit environment-specific fitness

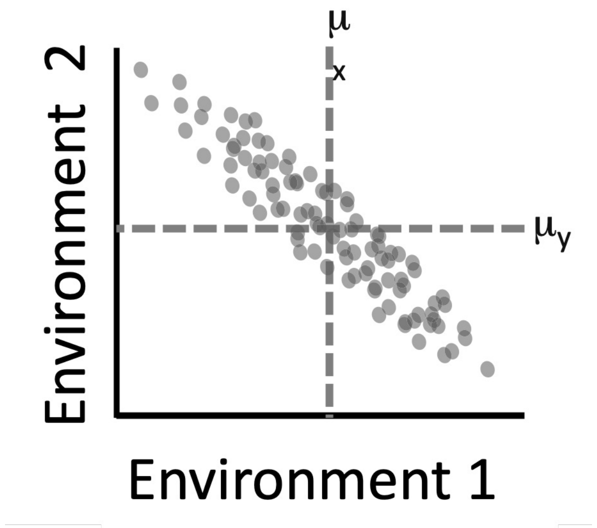




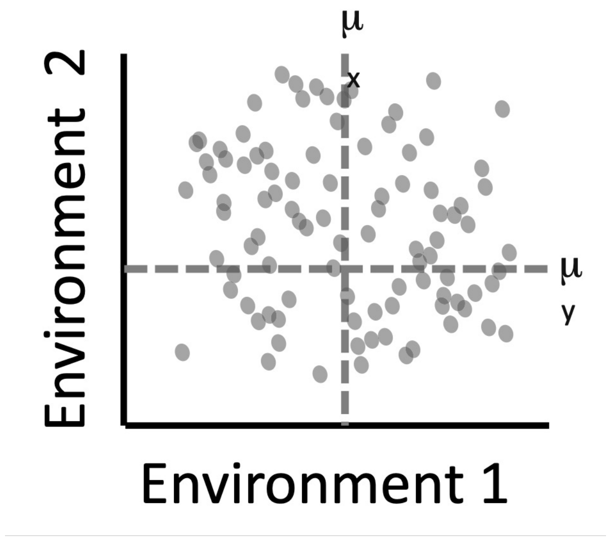
Burghardt et al. 2018, *PNAS*

Fitness correlations across environments provide clues about evolution

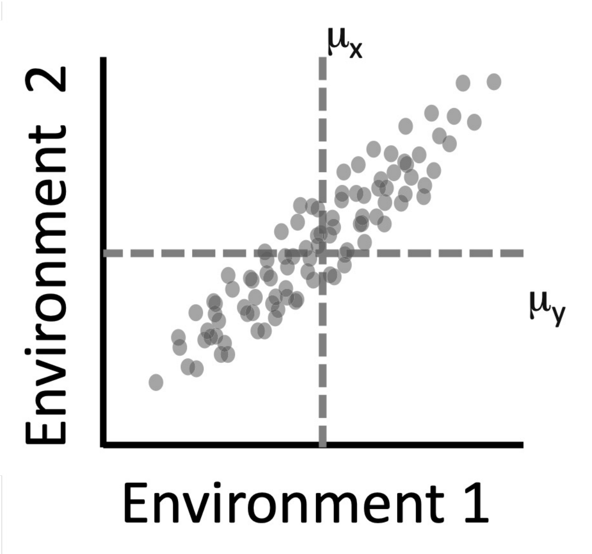
Constraint



Independence

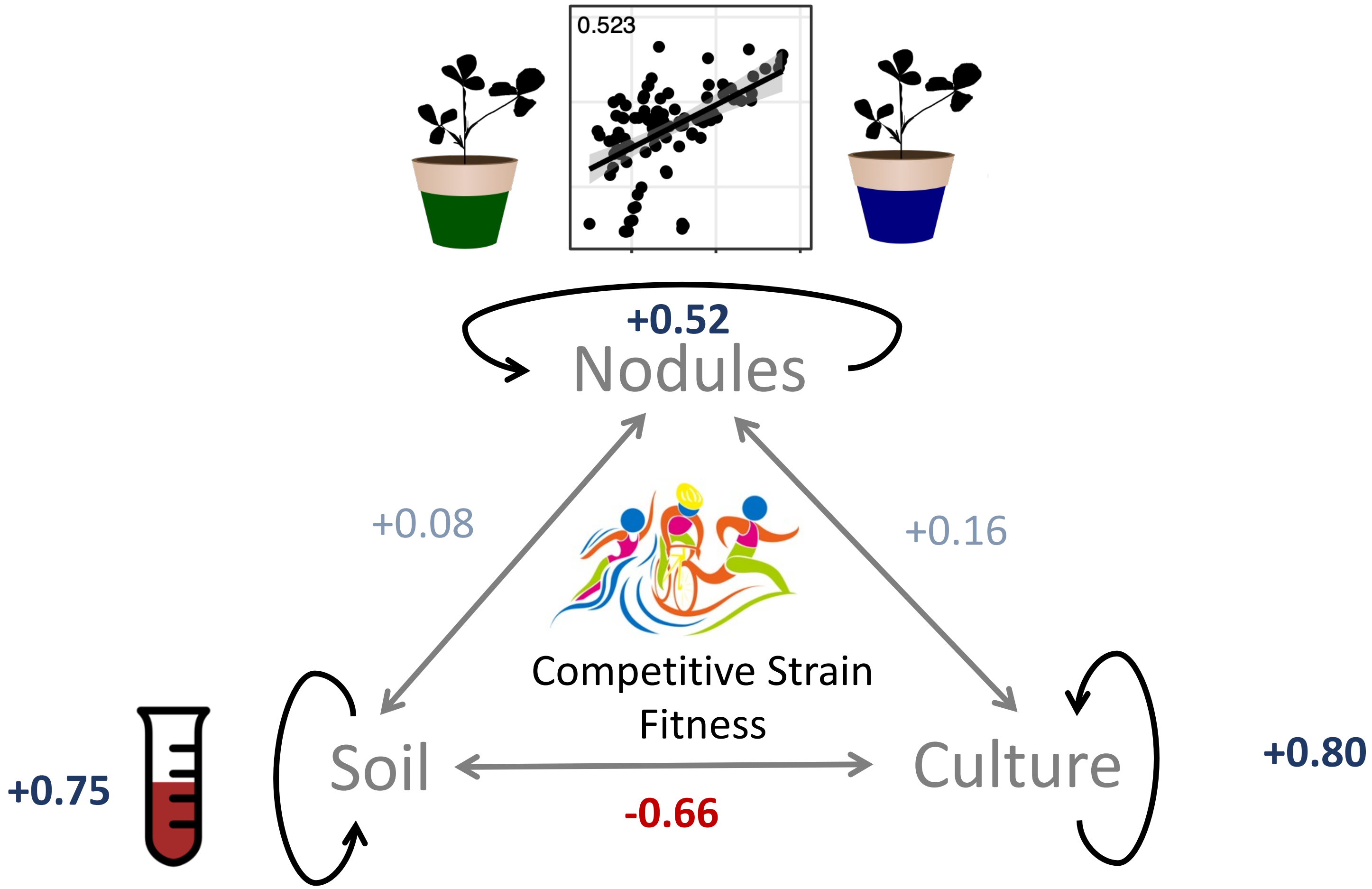


Facilitation



(Graphs: Salz et. al. 2017)

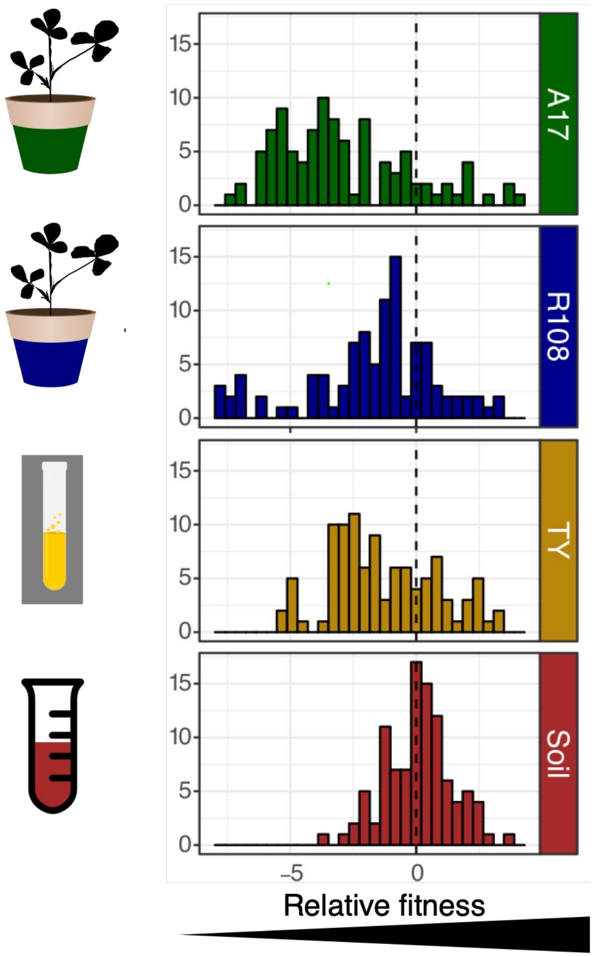
Fitness correlations across environments



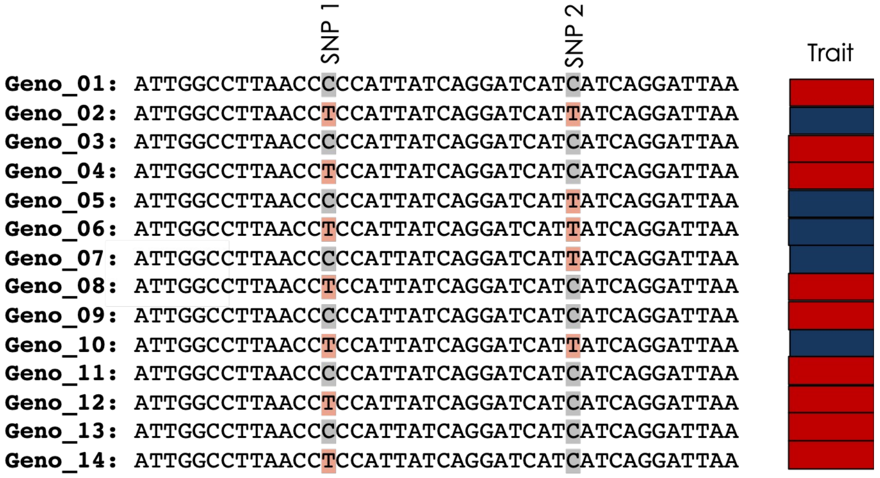
Burghardt et al. 2018, *PNAS*

Select & Resequence data allow downstream analysis

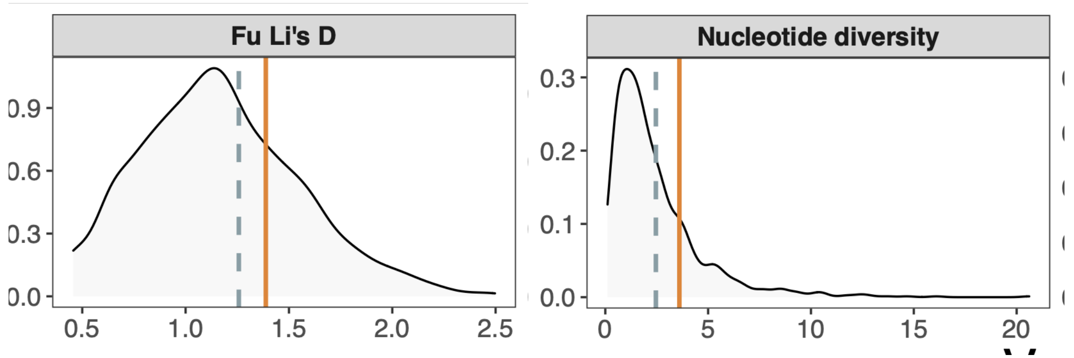
1. Measure Fitness



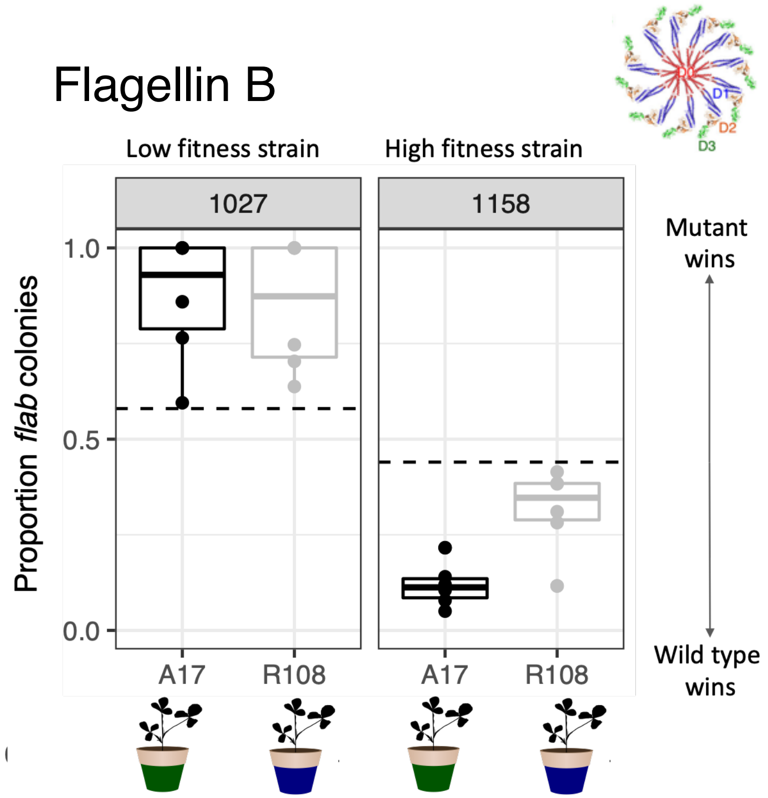
2. Genome wide association studies



3. Population Genetic Analysis



4. Validate Candidate Genes



Burghardt *PNAS* 2018
Epstein *mSphere* 2018
Burghardt *Evolution* 2019
Burghardt 2020 *Plant Physiology*
Batstone et.al 2022 *Proc B*
Burghardt et al 2022 *AEM*
Epstein… Tiffin 2022 *Molecular Ecology*

Can use GWAS tools in BOTH hosts simultaneously

FROM THE COVER

MOLECULAR ECOLOGY WILEY

Combining GWAS and population genomic analyses to characterize coevolution in a legume-rhizobia symbiosis

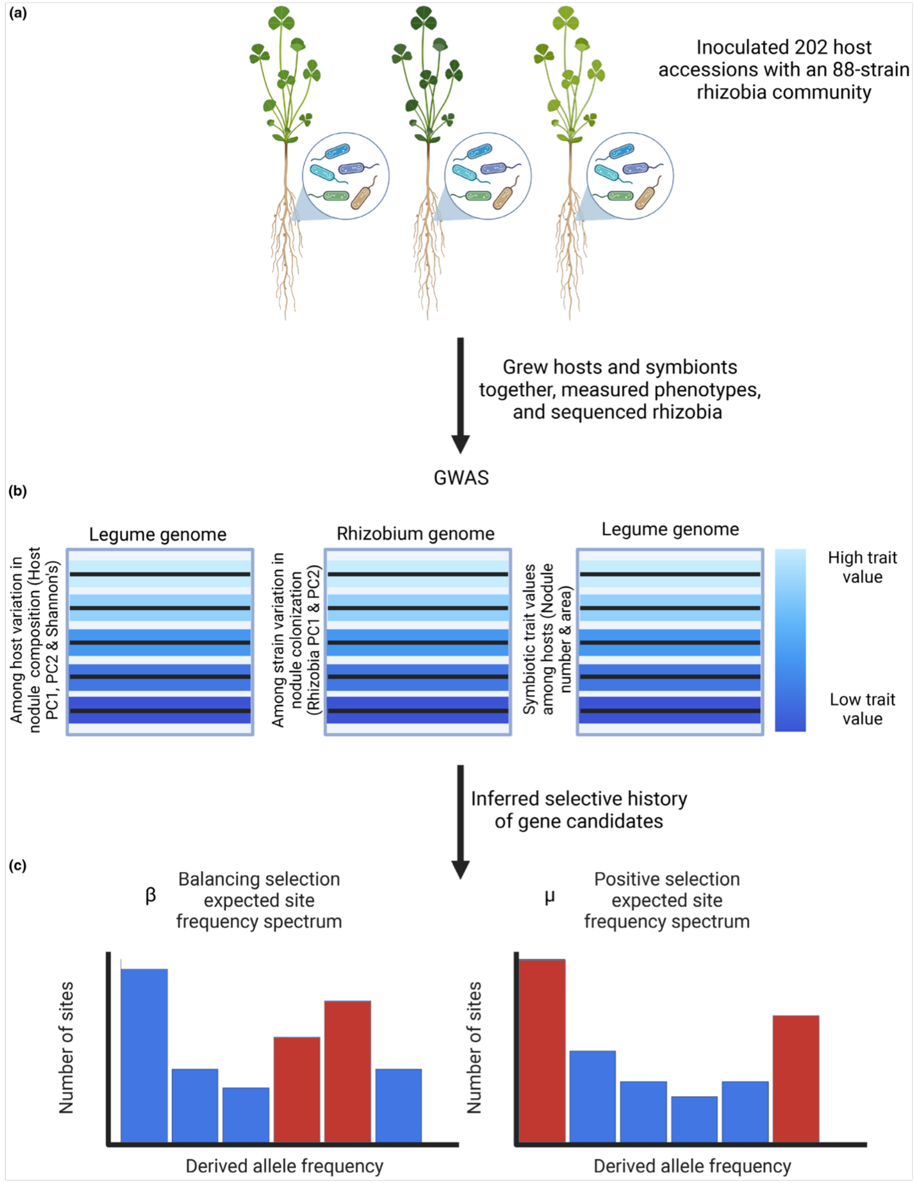
Brendan Epstein[1] | Liana T. Burghardt[2] | Katy D. Heath[3,4] | Michael A. Grillo[5] | Adam Kostanecki[1] | Tuomas Hämälä[1] | Nevin D. Young[1,6] | Peter Tiffin[1]

PERSPECTIVE

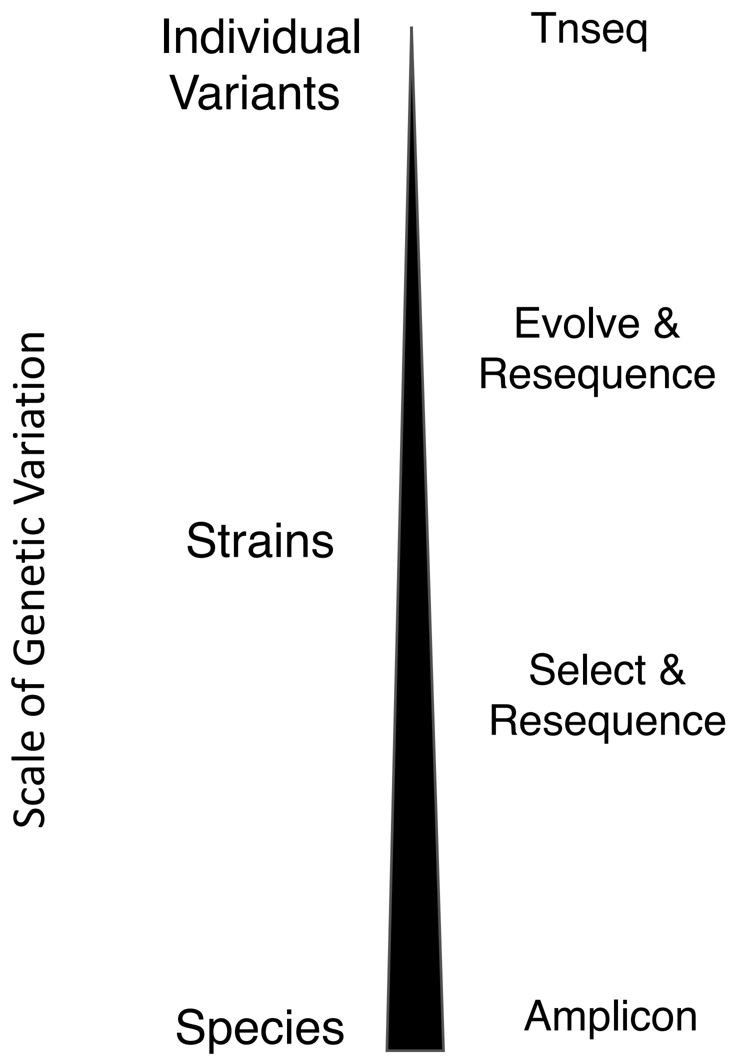
MOLECULAR ECOLOGY WILEY

An emerging view of coevolution in the legume–rhizobium mutualism

Christopher Carlson | Megan E. Frederickson



Whole genome shotgun sequencing allows measurement of microbial symbiont fitness at many different levels of genetic variation



ADAPTATION

Experimental evolution makes microbes more cooperative with their local host genotype

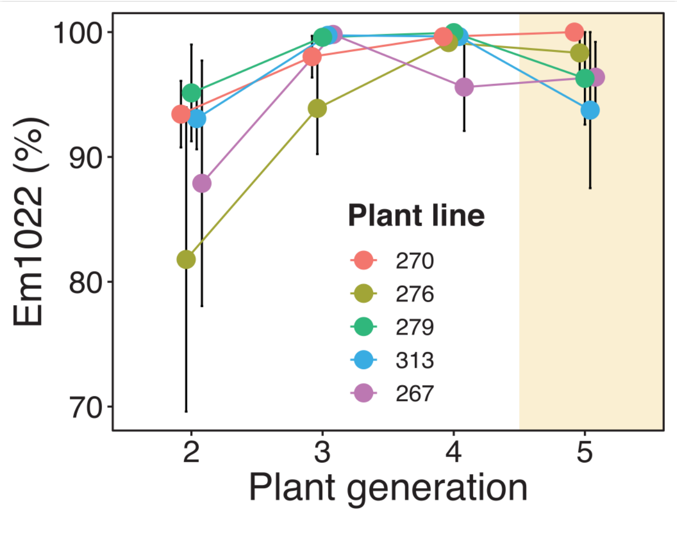
Rebecca T. Batstone[1,2*], Anna M. O'Brien[1,3], Tia L. Harrison[1], Megan E. Frederickson[1,4,5]

PNAS

Experimental evolution makes microbes more cooperative with their local host genotype

Science

1. Evolve two strains for 5 host generations on 5 host genotypes



High quality microbial partner spread to near fixation

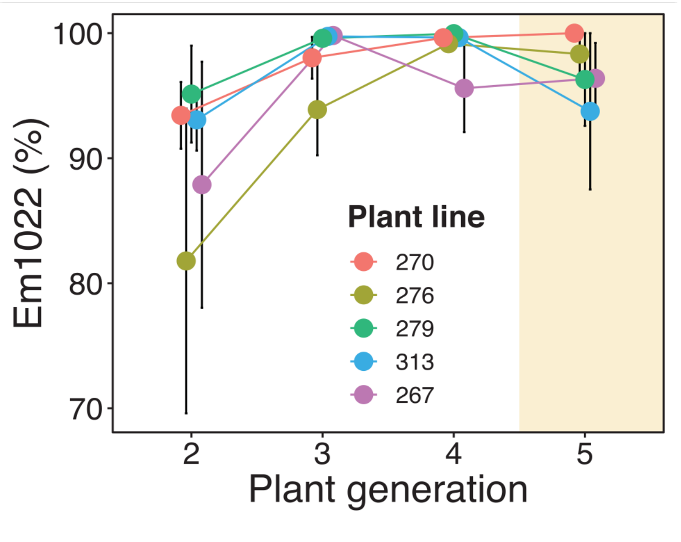
Experimental evolution makes microbes more cooperative with their local host genotype

Science

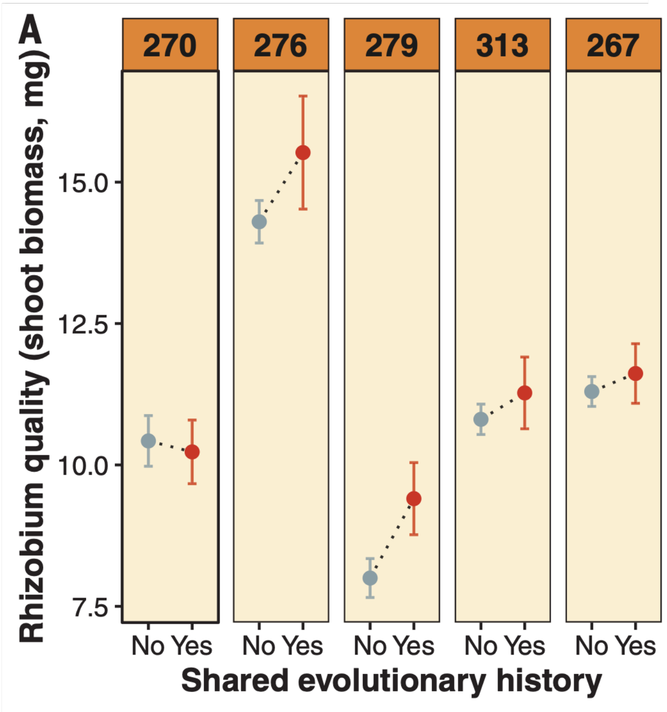
1. Evolve two strains for 5 host generations on 5 host genotypes

2. Isolate evolved strains and sequence

3. Assess phenotypic consequences of evolution



High quality microbial partner spread to near fixation



Isolates are most helpful to the hosts they evolve on

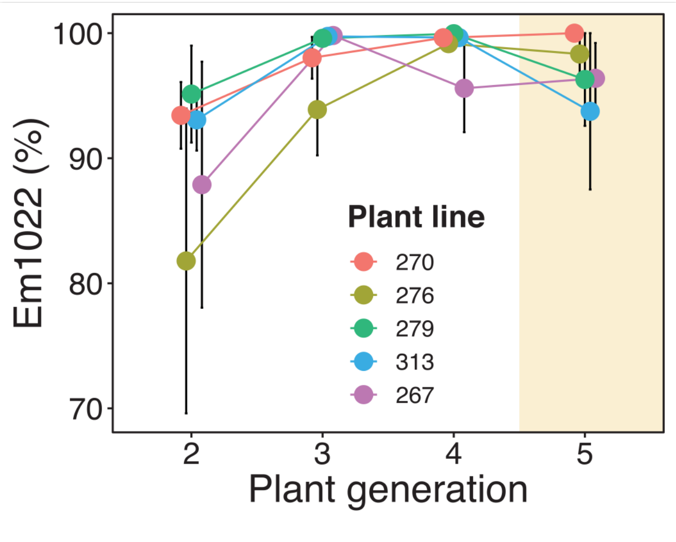
Experimental evolution makes microbes more cooperative with their local host genotype

1. Evolve two strains for 5 host generations on 5 host genotypes

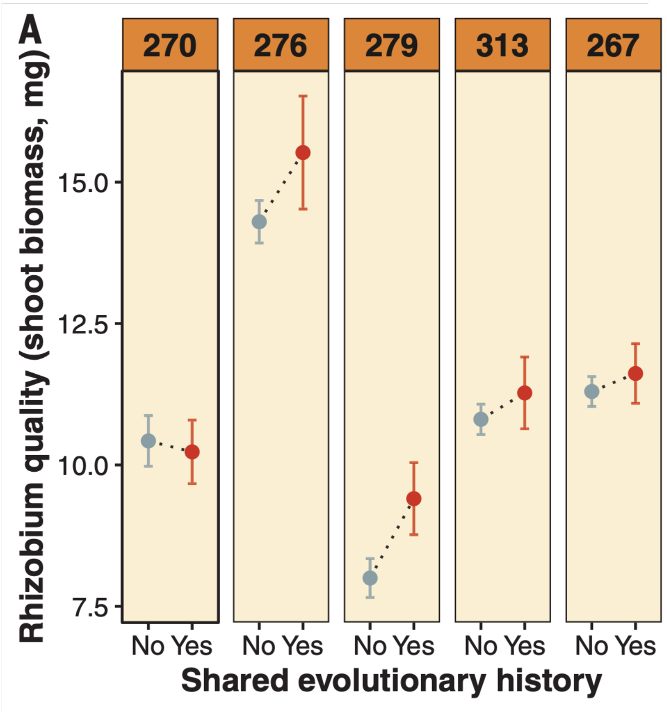
2. Isolate evolved strains and sequence

3. Assess phenotypic consequences of evolution

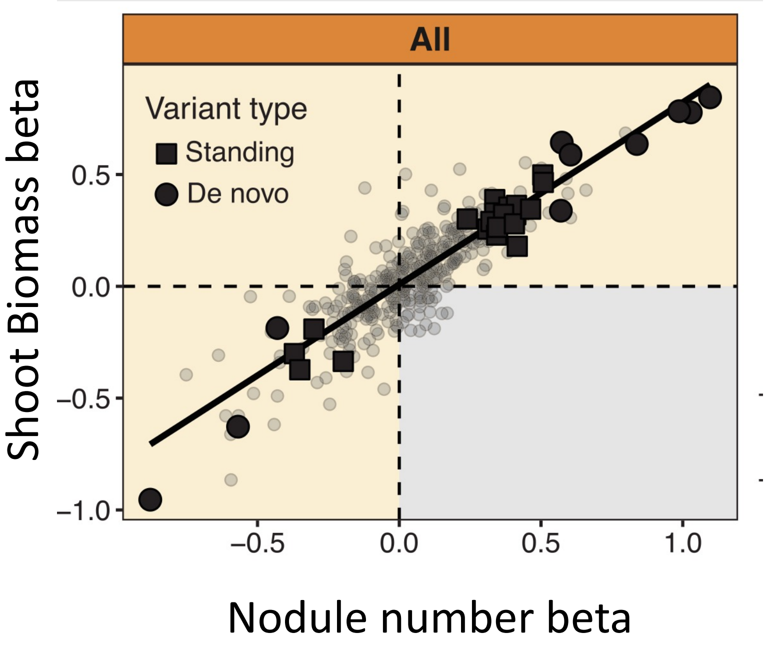
4. Assess genomic patterns of evolution in response to hosts



High quality microbial partner spread to near fixation

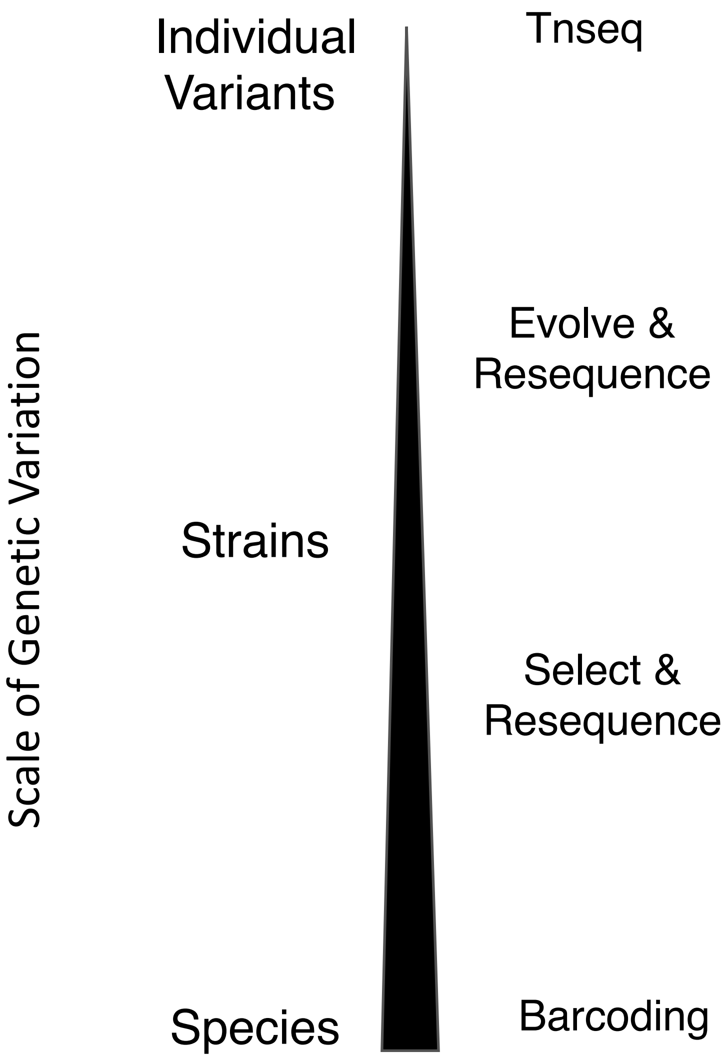


Isolates are most helpful to the hosts they evolve on



Both standing and de novo genetic variation helped both partners

Whole genome shotgun sequencing allows measurement of microbial symbiont fitness at many different levels of genetic variation



PNAS

Lifestyle adaptations of *Rhizobium* from rhizosphere to symbiosis

Rachel M. Wheatley[a,1], Brandon L. Ford[a,1], Li Li[b,1], Samuel T. N. Aroney[a], Hayley E. Knights[a], Raphael Ledermann[a], Alison K. East[a], Vinoy K. Ramachandran[a,2], and Philip S. Poole[a,2]

Science

Transposon and insertion sequencing is increasingly used to study symbiosis...

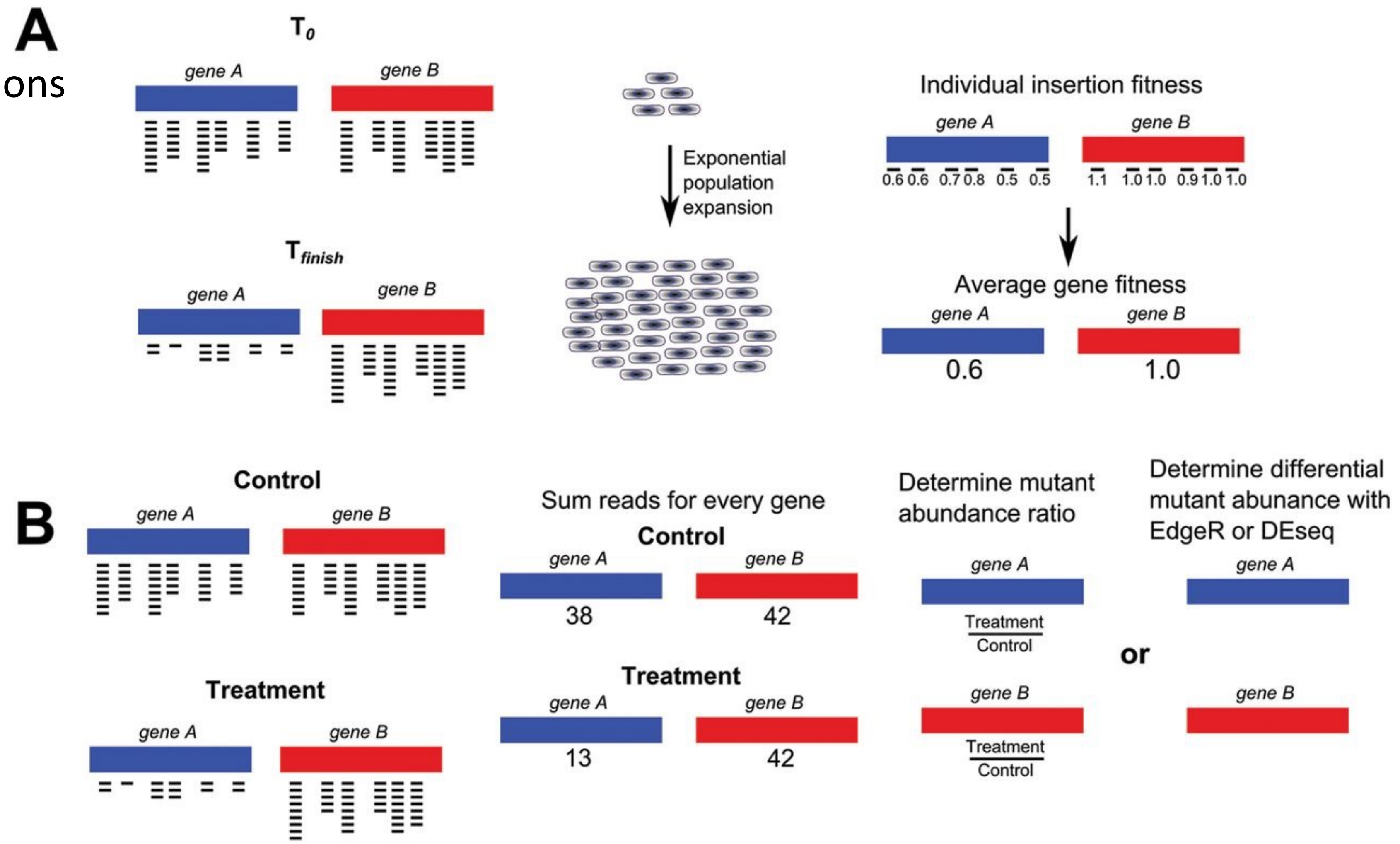
1. Create library filed with mutations

2. Expose to selection

3. Whole genome sequencing

Inference: if genetic disruptions stick around... the gene is not essential in that environment

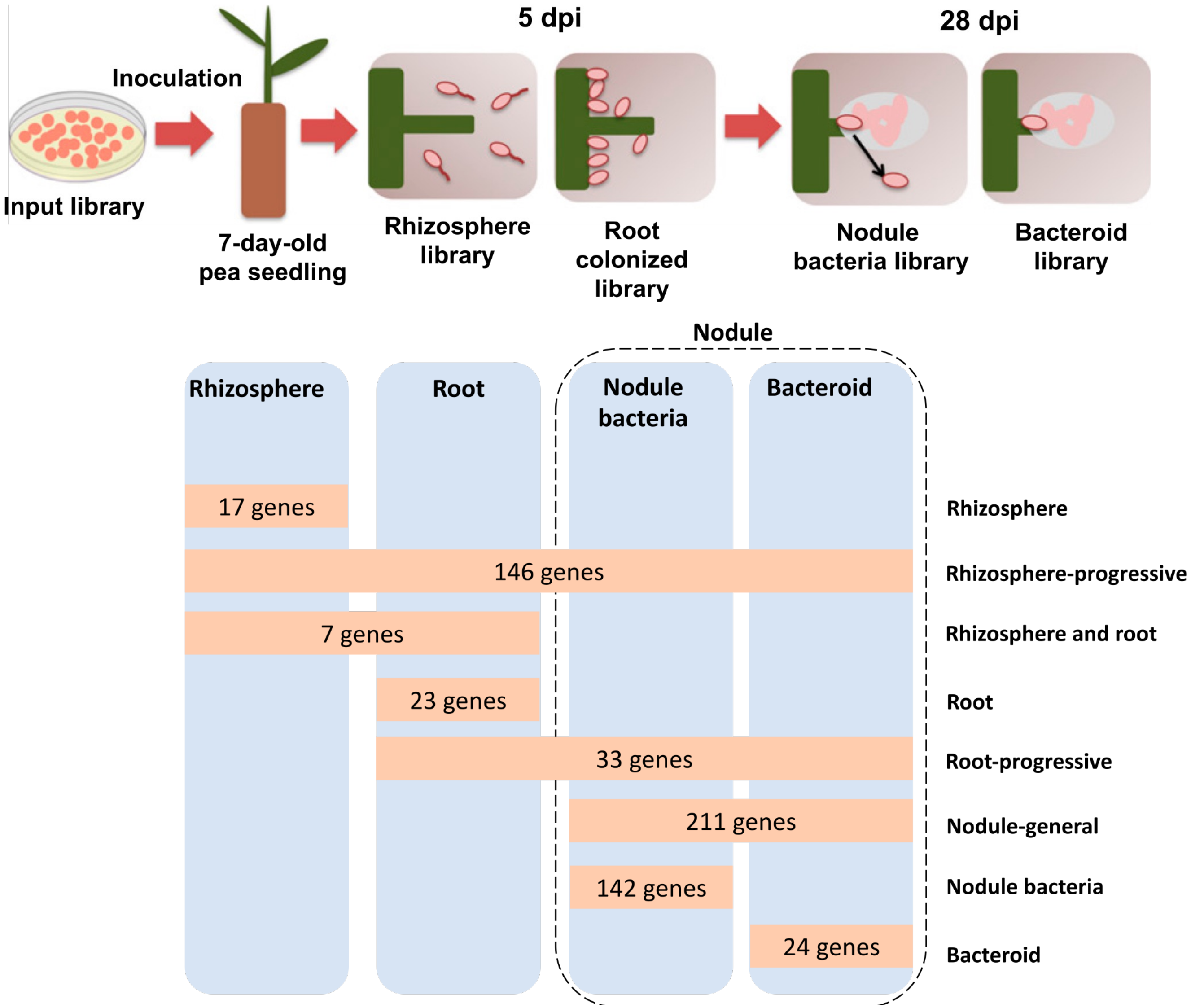
Limitations: Only screens knockout disruptions, not quantitative variation + can't have huge bottlenecks



Implementation and Data Analysis of Tn-seq, Whole-Genome Resequencing, and Single-Molecule Real-Time Sequencing for Bacterial Genetics

Authors: Peter E. Burby, Taylor M. Nye, Jeremy W. Schroeder, Lyle A. Simmons

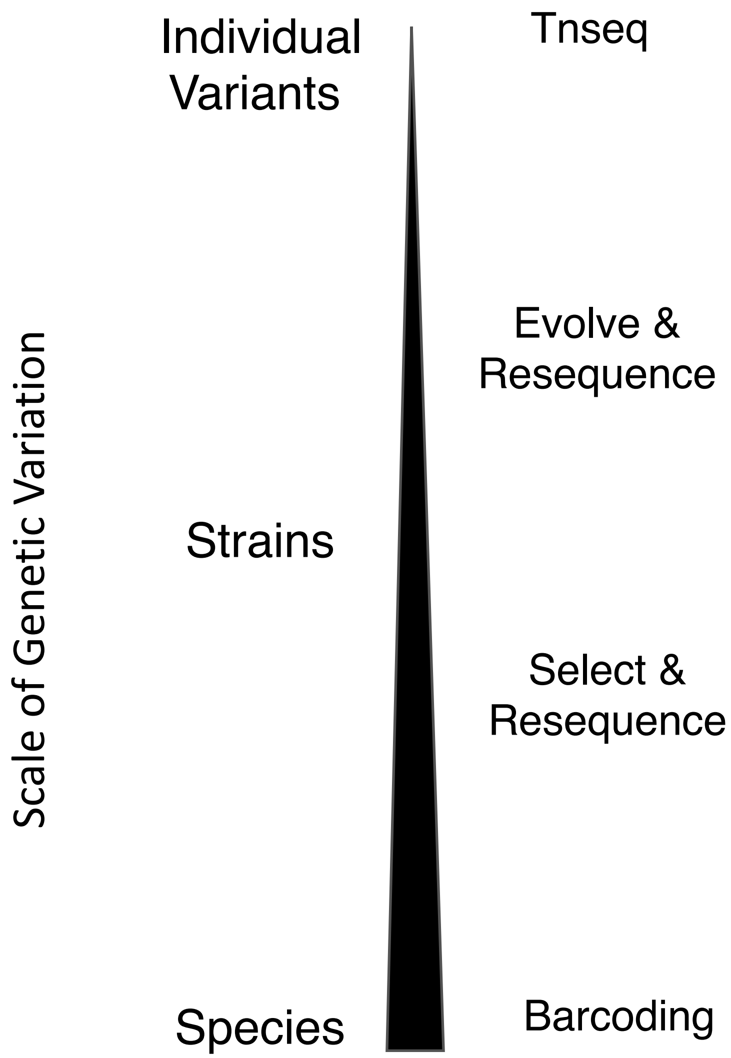
Transposon/Insertion Sequencing: Lifestyle adaptations of *Rhizobium* from rhizosphere to symbiosis



- Categorize rhizobial genes with essential functions across one or more host created environments

Wheatley, PNAS

Much cheaper if you can use genome engineering to add unique DNA barcodes to strains or mutant libraries!



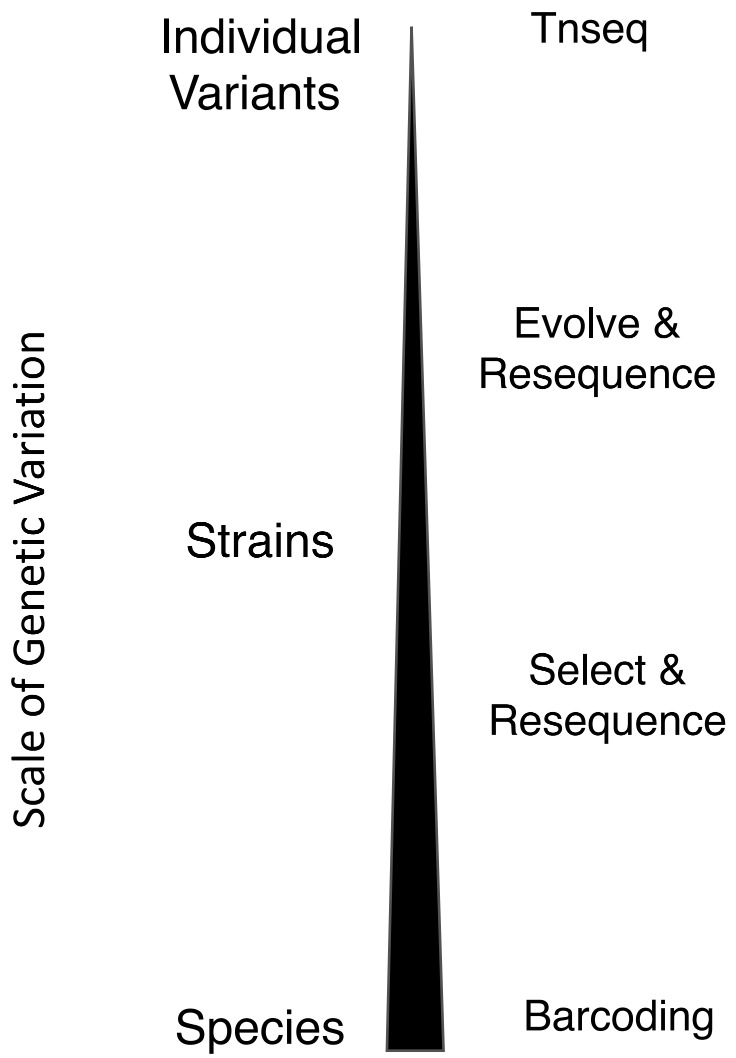
Rapid Quantification of Mutant Fitness in Diverse Bacteria by Sequencing Randomly Bar-Coded Transposons

Kelly M. Wetmore,[a] Morgan N. Price,[a] Robert J. Waters,[b] Jacob S. Lamson,[a] Jennifer He,[c] Cindi A. Hoover,[b] Matthew J. Blow,[b] James Bristow,[b] Gareth Butland,[c] Adam P. Arkin,[a,d] Adam Deutschbauer[a]

Adaptation of Plasmid-ID Technology for Evaluation of N_2-Fixing Effectiveness and Competitiveness for Root Nodulation in the *Sinorhizobium–Medicago* System

Jake D. Schumacher[1] | Nicholas Dusek[2] | Marcela Mendoza-Suárez[3] | Barney A. Geddes[1]

Whole genome shotgun sequencing allows measurement of microbial symbiont fitness at many different levels of genetic variation



PNAS

Lifestyle adaptations of *Rhizobium* from rhizosphere to symbiosis

Rachel M. Wheatley[a,1], Brandon L. Ford[a,1], Li Li[b,1], Samuel T. N. Aroney[a], Hayley E. Knights[a], Raphael Ledermann[a], Alison K. East[a], Vinoy K. Ramachandran[a,2], and Philip S. Poole[a,2]

ADAPTATION

Science

Experimental evolution makes microbes more cooperative with their local host genotype

Rebecca T. Batstone[1,2]*, Anna M. O'Brien[1,3], Tia L. Harrison[1], Megan E. Frederickson[1,4,5]

PNAS

Select and resequence reveals relative fitness of bacteria in symbiotic and free-living environments

Liana T. Burghardt[a], Brendan Epstein[a], Joseph Guhlin[a], Matt S. Nelson[b], Margaret R. Taylor[b], Nevin D. Young[a,c], Michael J. Sadowsky[a,b,d], and Peter Tiffin[a,1]

ISME

Variation in rhizosphere microbial communities and its association with the symbiotic efficiency of rhizobia in soybean

Qin Han[1] · Qun Ma[1] · Yong Chen[1] · Bing Tian[1] · Lanxi Xu[1] · Yang Bai[2] · Wenfeng Chen[3] · Xia Li[1]

Learning objectives

- Part 1: Setting the stage for symbiosis
  - Define symbiosis and understand variation in usage
  - Understand basic categories of symbiosis and brainstorm examples
  - Introduce pairwise vs. holobiont/hologenome frameworks

- Part 2: Leveraging sequencing to study symbiosis
  - Articulate how to use shotgun sequencing to measure symbiont fitness at multiple scales of genetic variation
  - Fall in love with legumes and rhizobia

"Our task now is to resynthesize biology; put the organism back into its environment; connect it again to its evolutionary past; and let us feel that complex flow that is organism, evolution, and environment united."-

Carl Woese 2003

Thank you!

Questions?

lianaburghardt@psu.edu
lianaburghardtlab.com

Image courtesy of the Soil and Terrestrial Environmental Physics group of Dani Or at ETH Zurich (www.step.ethz.ch). SEM image acquisition and coloring by Anne Greet Bittermann, ETH Zurich – ScopeM